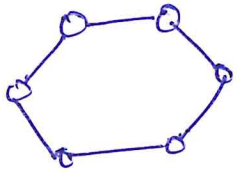
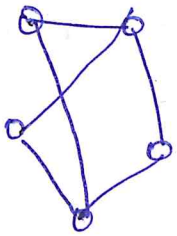
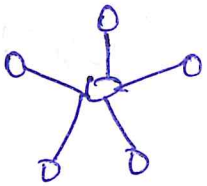
Network Topologies:



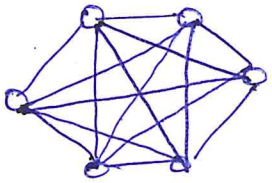
Ring



Mesh.



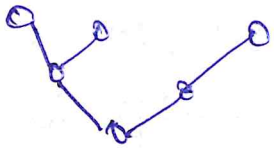
Star



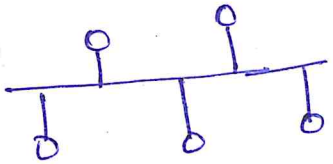
Fully connected



Line



Tree

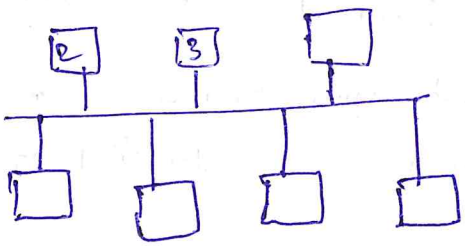


Bus

DLL

DATA LINK LAYER

1. Access to media (fiber optic, wifi, twisted cable)
2. Media-access-control



* It is going to control which pair of media will transfer the data.

3. Error-detection → If any collision then we have to detect the collision.

* Frame is the unit for data link layer.

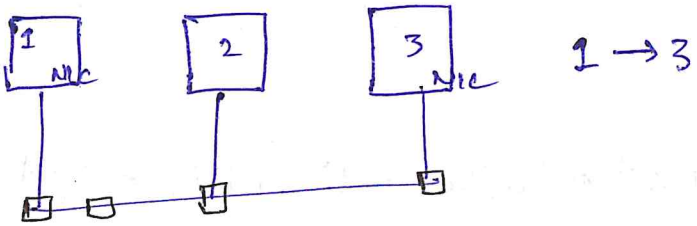
(4) Flow control → Ensures that two hosts will control at what rate data should be transfer so that there will be no. errors.

DLL

Logical link sublayer
flow control, ACK and error notification

Media-access-control sublayer
MAC address (wireshark)
[12 digit alpha numeric]
→ who access the media

Hardware: NIC ; WIFI-card ; ....
→ Available in your system.
Layer-2-Switch



* MAC address : → 48 bit address. [Unique]
* In wireshark you can see a UDP packet where that packet will be showing mac address.

* Switches, layer 2 switch used for connecting hosts.

H — S — H
↳ Amplify the signals also.

Delays :-

① Transmission delay (T_t)

How much time it will take to place the data on the physical wire :-

B = Bandwidth in bps L : # bits in a packet.

$$T_t = \frac{L}{B} \text{ sec}$$

- B bits → 1 sec.
- 1 bit → $1/B$ sec.
- L → L/B sec

② Propagation delay :- (T_p)

D : length of wire.

V : velocity

Propagation time is how much time it will take to send the data from one point to another point.

$$T_p = \frac{D}{V} \text{ sec}$$

* Eg D = $2 \cdot 10^3$ KM

V = 70% of speed in vacuum $V = 3 \times 10^8$ sec.

$$T_p = \frac{2100}{2 \cdot 1 \times 10^8} \cdot \text{sec} = 10^{-5} \text{ sec} = 10 \mu\text{sec}$$

Note:→

$$\begin{cases} 1\ ms = 10^{-3}\ sec \\ 1\ \mu s = 10^{-6}\ sec \\ 1\ ns = 10^{-9}\ sec \end{cases}$$

Data:
$1K = 2^{10} = 1024$

Bandwidth:
$1K = 10^3$

↓
Typical convention.

Eg. $L = 1000$ bits

$B = 1$ Kbps

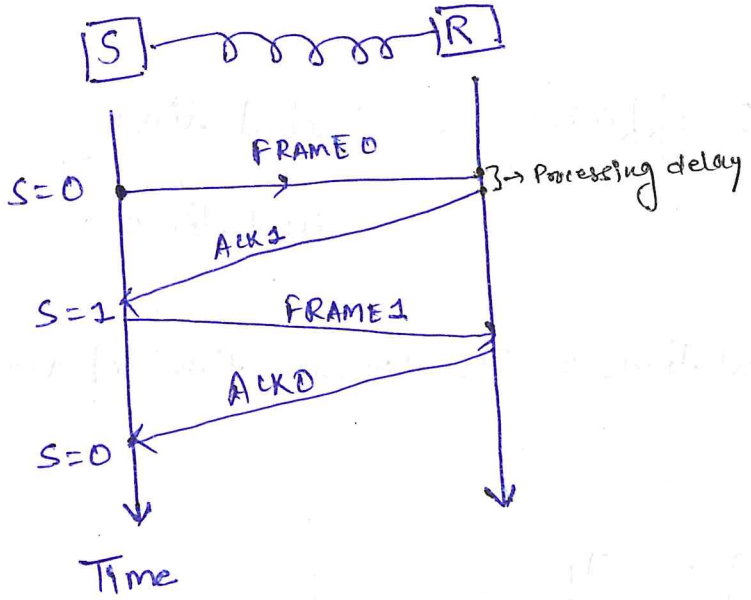
$$T_t = \frac{1000 \text{ bits}}{1 \text{ Kbps}} = \frac{1000}{1000} = 1 \text{ sec}$$

Eg $L = 1$ Kb

$B = 1$ Kbps

$$T_t = \frac{1024}{1000} = 1.024 \text{ sec}$$

Flow control: STOP and WAIT (ARQ): — Automatic repeat request



The flow is like 0, 1 0 1 0 1 . . . .

→ Here in Stop and wait, the sender send the frame zero and receiver send the Ack 1.

→ The sender waits until the Ack received from the receiver.

Calculation of total time for sending 1 packet / frame.

1 frame = $T_t + T_p + (T_{pr}) + (T_t^a) + T_p$

T_t^a → v.v small

T_{pr} → processing time is negligible

$(L) \rightarrow T_t + 2T_p$ sec

→ Total time to send a frame.

L → L bits

Calculation of "Efficiency". -

η = efficiency = $\frac{\text{Useful time}}{\text{Total time}}$.

→ Useful time = Transmission time of frame.

* $\boxed{\eta = \frac{T_t}{(T_t + 2T_p)}}$ — (1)

* Let's divide the eq 1 with T_t in both numerator and denominator.

$$\eta = \frac{1}{1 + 2(T_p/T_t)}, \quad \boxed{a = \frac{T_p}{T_t}}$$

* $\boxed{\eta = \frac{1}{1+2a}}$

Throughput / Bandwidth-utilization / Effective Bandwidth.

* Throughput is measured on bits per seconds.

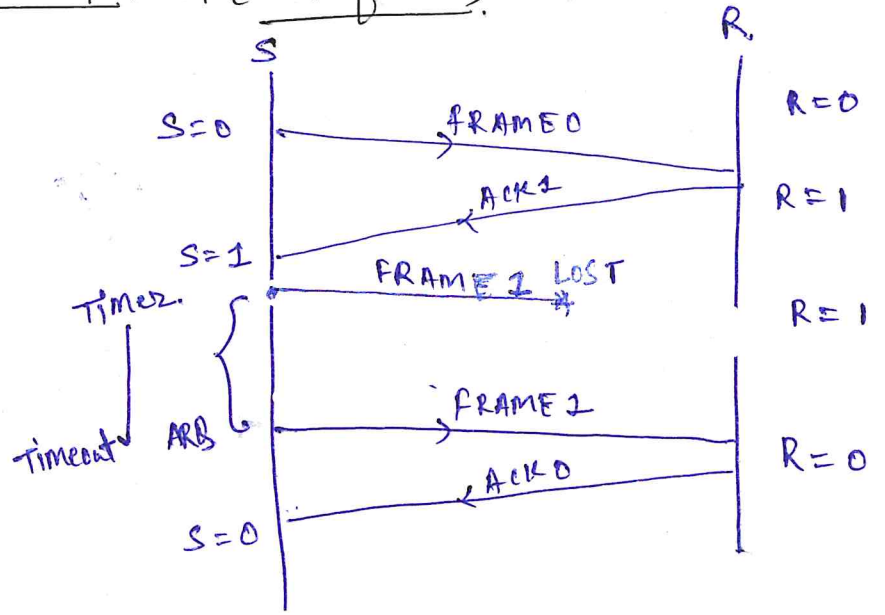
* **Throughput = $\frac{L}{T_t + 2T_p}$** ⇒ $\frac{\text{Datasize}}{\text{Total time}}$

$= \frac{\frac{L}{B} \times B}{T_t + 2T_p} = \frac{T_t \times B}{(T_t + 2T_p)} = \eta \times B$

* **Throughput = $\eta \times B$**

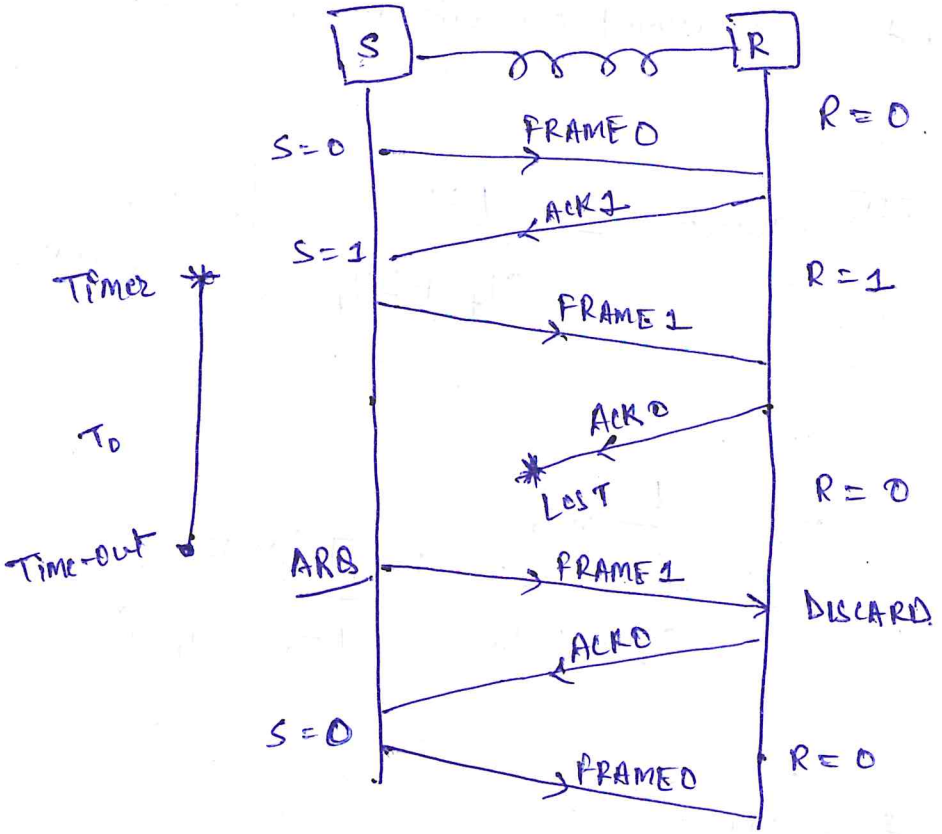
* In total time how many bits you send is called throughput.

STOP and wait (lost frame)



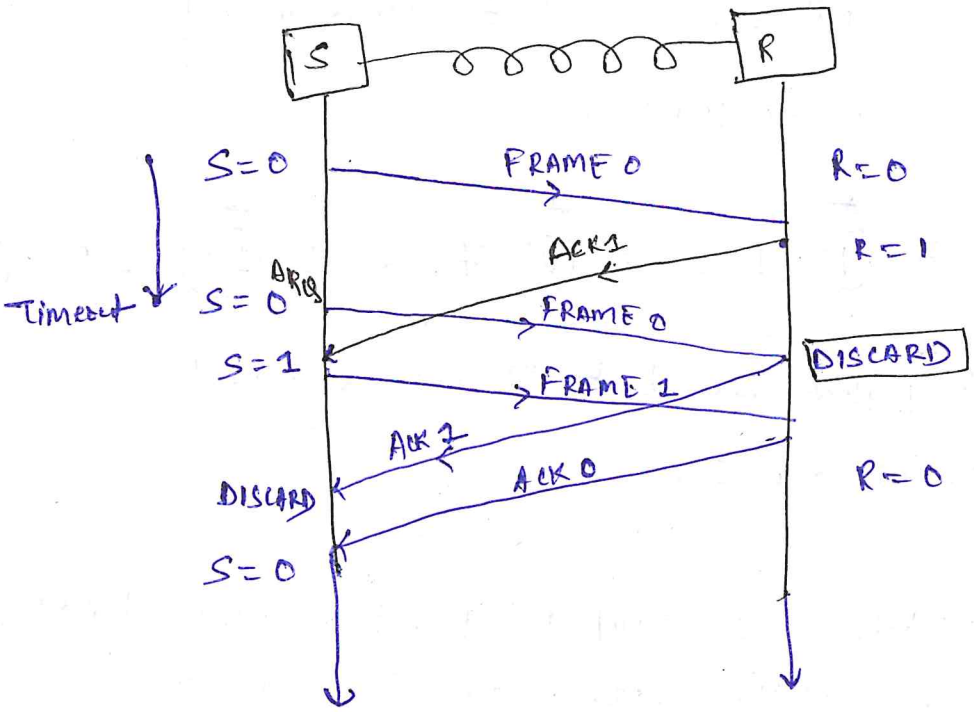
* Here when the frame get lost after some time when timer expires. It sends the same again.

STOP & wait ARQ (Lost Ack- frame)



* Here when the Ack get lost. So after timeout Sender will re-send that frame. At the receiving end receiver find out it's a duplicate frame and simply discard that frame. Receiver now send the Ack again.

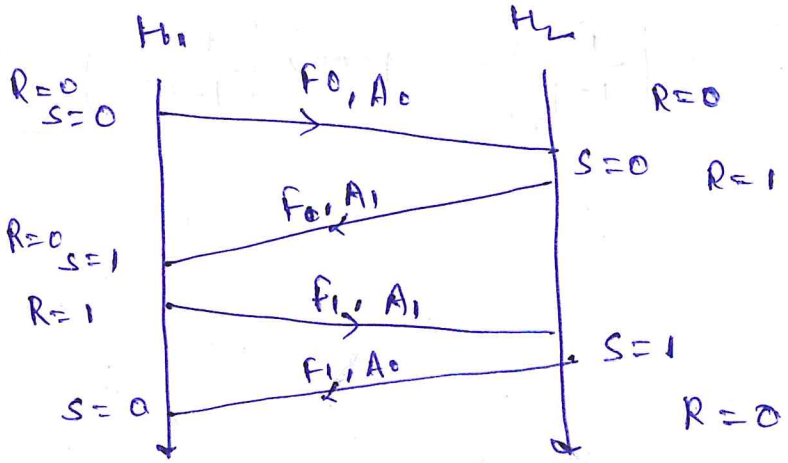
STOP and WAIT (delayed ACK)



In stop and wait when Ack get's delayed before that timeout occurs on the sender side and sender re-send the frame again. After sending the frame the sender's receive and Ack of the old frame and after some time again get's the ACK of the FRAME 0.

Sender's discard the Ack 1 because it is a duplicate ACK.

STOP and wait (Piggybacking) :-



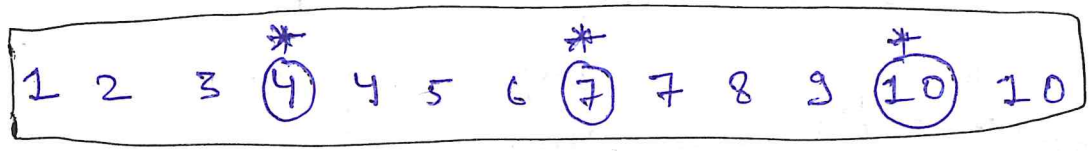
* Piggybacking means with data we also send the Ack together to save the time.

*. Here we are sending data to both the directions.

* Both hosts are sending data and receiving Ack synchronously

Que In Stop and wait ARQ, a sender sends 10 packets of which every 4th packet is lost. How many transmissions are needed?

Soln →



Every 4th packet is lost it means total Transmissions are 13.

Que: If 1000 packets are send using stop and wait ARQ. The error probability is 80%. No. of transmissions needed?

Soln: 80% probability $= \frac{1000 \times 80}{100} = 800$ packets

$= \frac{800 \times 80.1}{100} = 640 \ldots\ldots$

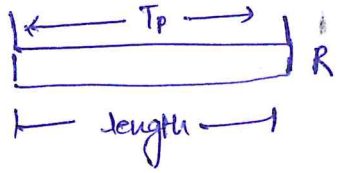
So, $n + np + np^2 + \ldots\ldots \rightarrow$ G.P.

$\Rightarrow n[1 + p + p^2 + p^3 \ldots]$

$\Rightarrow n\left(\frac{1}{1-p}\right) = 1000 \times \left(\frac{1}{1-0.8}\right) = \frac{1000}{0.2} = 5000$ packets.

Capacity of a channel / medium: Maximum no. of bits on the wire.

$$= Bw \times T_p \text{ bits}$$



Cap ↑ ⇒ Bw ↑

Cap ↑ ⇒ T_p ↑ ⇒ length ↑

Que:



Bw = 2.1 Mbps

T_p = 1 sec.

frame size = 10^3 bits

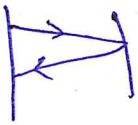
Capacity = ? capacity (full duplex) = ?

Soln. Capacity = $2.1 \times 10^6 \times 1$ sec $= 2.1 \times 10^6$ b $= 2100$ frames.

Capacity full duplex = 2100 + 2100 = 4200 frames.

In Stop and wait ARQ

$$\text{Efficiency } (\eta) = \frac{1}{4200}.$$

In Stop and wait protocol we are sending only one packet in round trip time. RTT means  $= 2T_p$.

*. η = No. of packets at any time on the medium / Capacity

Que. T_p = 1 ms T_t = 1 ms

No. of packets can be transmitted before waiting for an Ack of the 1st packet.

Soln. S ——〰〰〰—— R

It means capacity.

1 packet → T_t sec

$\frac{1}{T_t}$ packets ← 1 sec

Total time = $T_t + 2t_p$

In total time we can place = $\frac{T_t + 2t_p}{t_t}$ packets

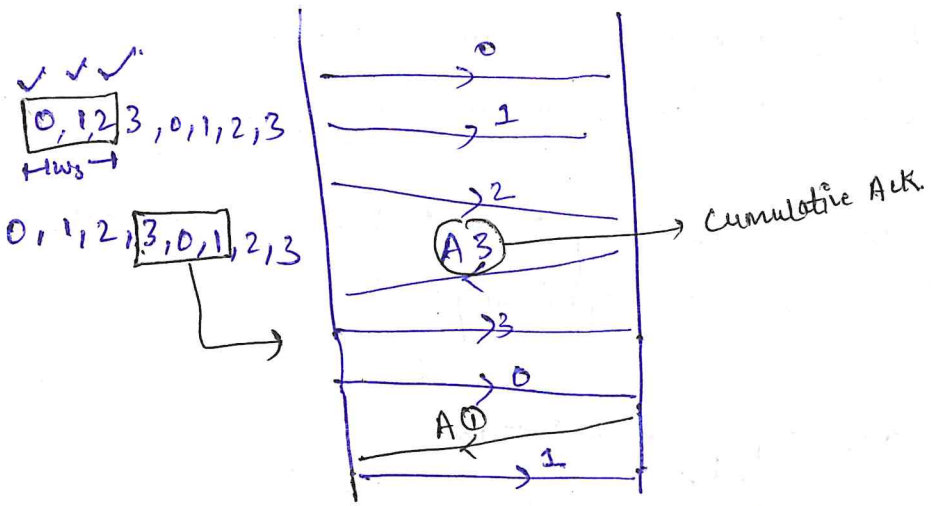
$[t_p/t_t = a]$ = $1 + 2a$ [max packets we can place]

= 3 packets (Ans)

Capacity = $1 + 2a$.

η = Stop and wait = $\frac{1}{1+2a}$

Sliding window protocols :-

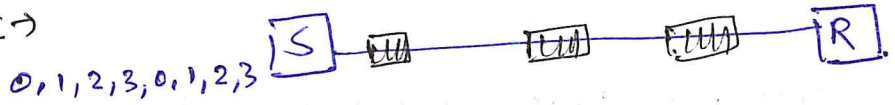


* Here we are using 2 or more than 2 bits for sequence number.

* Using 2 bits we can represent 4 numbers. = 0, 1, 2, 3

* The window size is decide on the basis of sequence bits.

Concept →



In m bits = 2^m sequence no. possible.

$ws \leq 2^m$ [because we can not repeat same seq no. again]

and

$(ws \leq 1+2a)$

$$\text{Max}^m\ ws \leq \min(2^m, 1+2a)$$

capacity = 1+2a packets.

* For 1+2a packets → $\log_2(1+2a)$ bits are required.

* min m = $\log_2(1+2a)$, max m = ∞

Que : $T_p = 49.5$ ms

$T_t = 1$ ms.

(A) what is the max-window size.

(B) No. of bits for sequence no.

(C) If we have only 5 bit for sequence no. what is the efficiency.

Soln (A) $a = T_p / t_t = 49.5$ ms

$ws \leq \min(2^M, 1+2a)$

$1 + 2 \times 49.5 = 100$

$ws \leq 100$

(B) $ws = 100$ So, No. of seq no $= \lceil \log_2 100 \rceil$

$= 7$ bits

(C) using 5 bits we can send only

$0, 1, 2, \cdots 31 = 32$ packets

$\eta = \frac{32}{100} = 32\%$

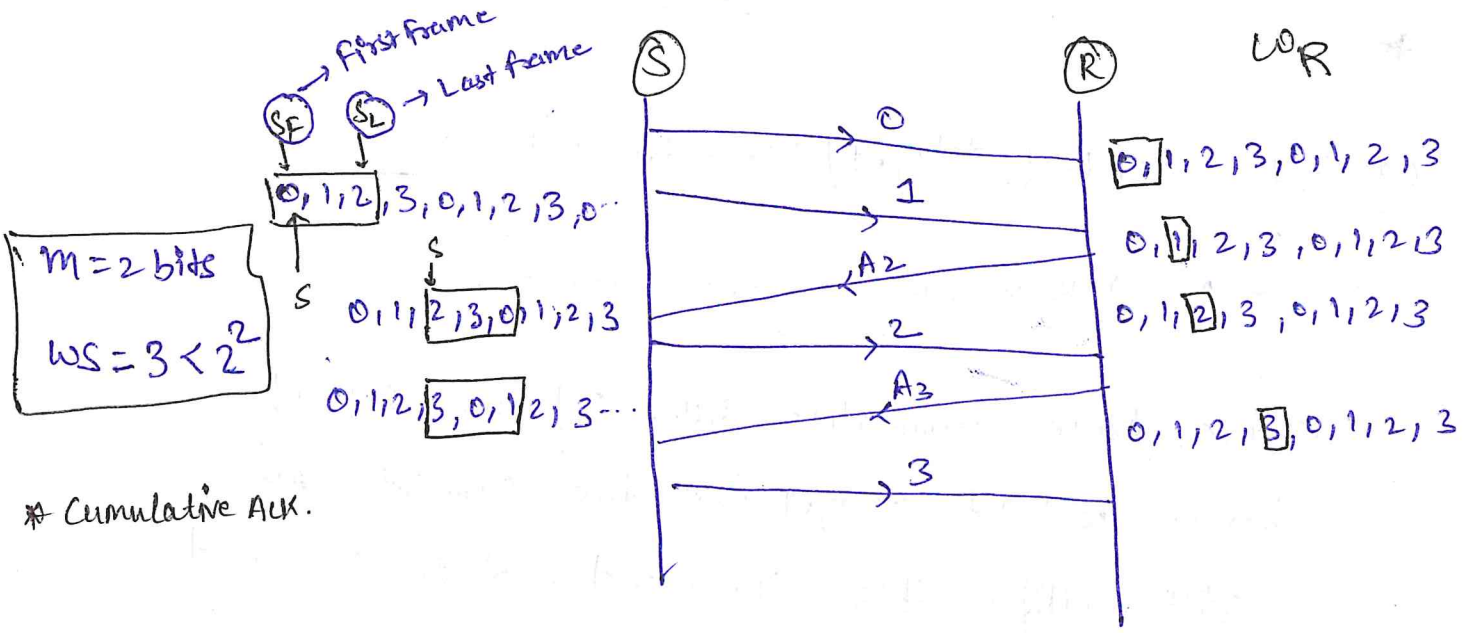
* So, Efficiency of Slidingwindow

$$\eta_{Sw} = \frac{w_s}{1+2a}.$$

Throughput $= \eta_{Sw} \times B$

Go-back-N ARQ

Sliding window



* Cumulative ACK.

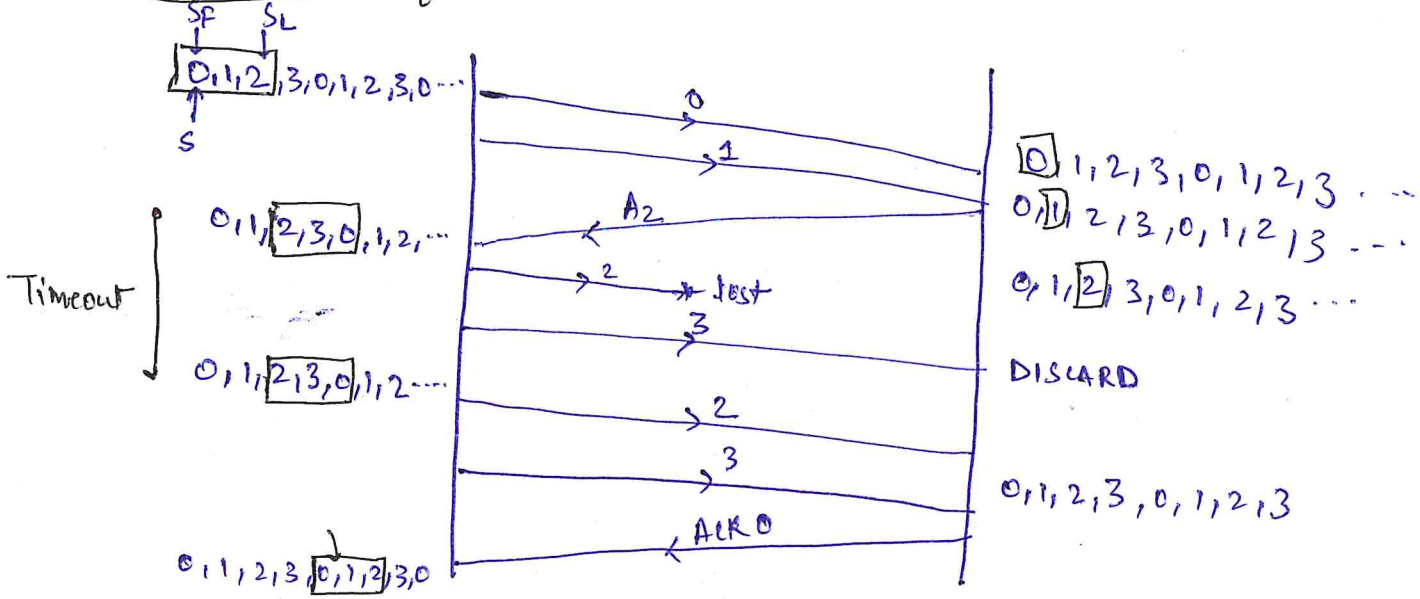
In Go Back N ARQ

$Sws \leq 2^m - 1$ m = seq no.

$Rws = 1$

GBN (Lost frame) :

N → width of ws



* If the frame is lost, we are going back N and send all the frames in the windows again.

Case-1 GBN (delayed / lost / damaged ACK)

↳ Ack arrives before Timeout.

* We have cumulative Ack. If any Ack of frame will get delayed then the cumulative Ack will nullify it's. We need not to re-send the frame again.

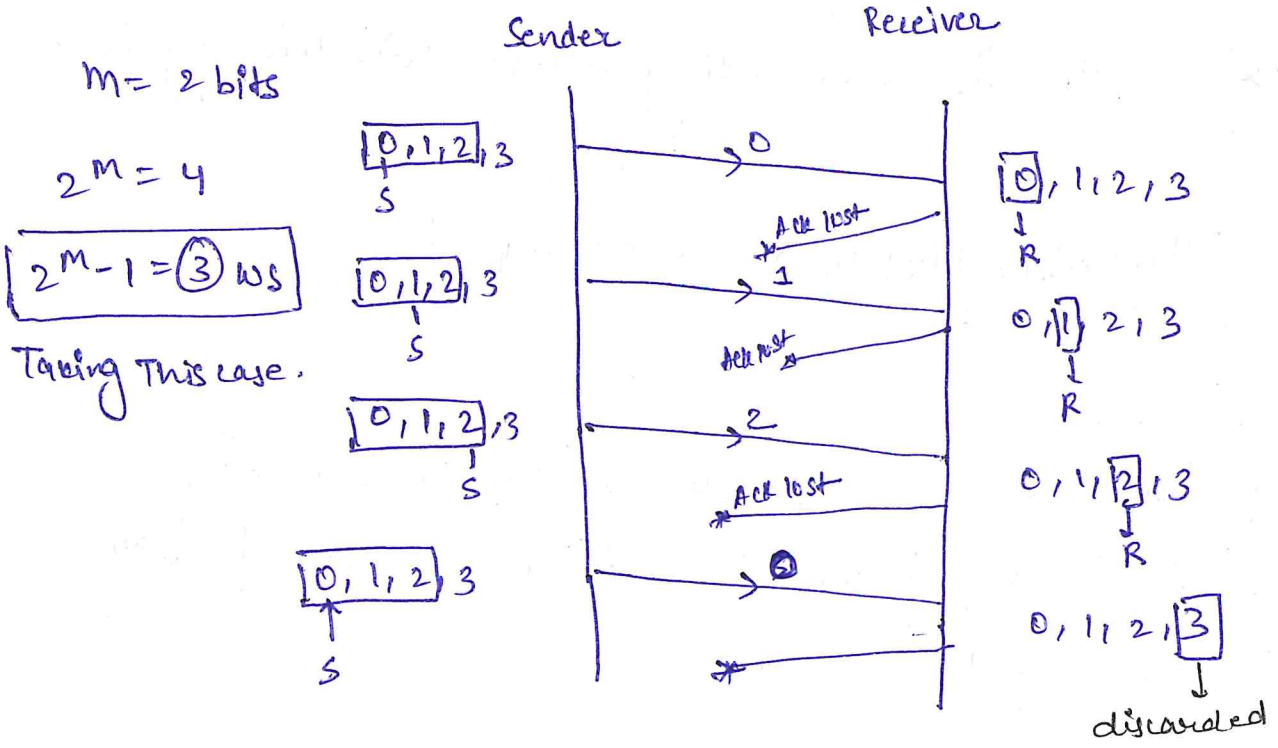
* Ack arrives after Timeout

⇓

resend all frames.

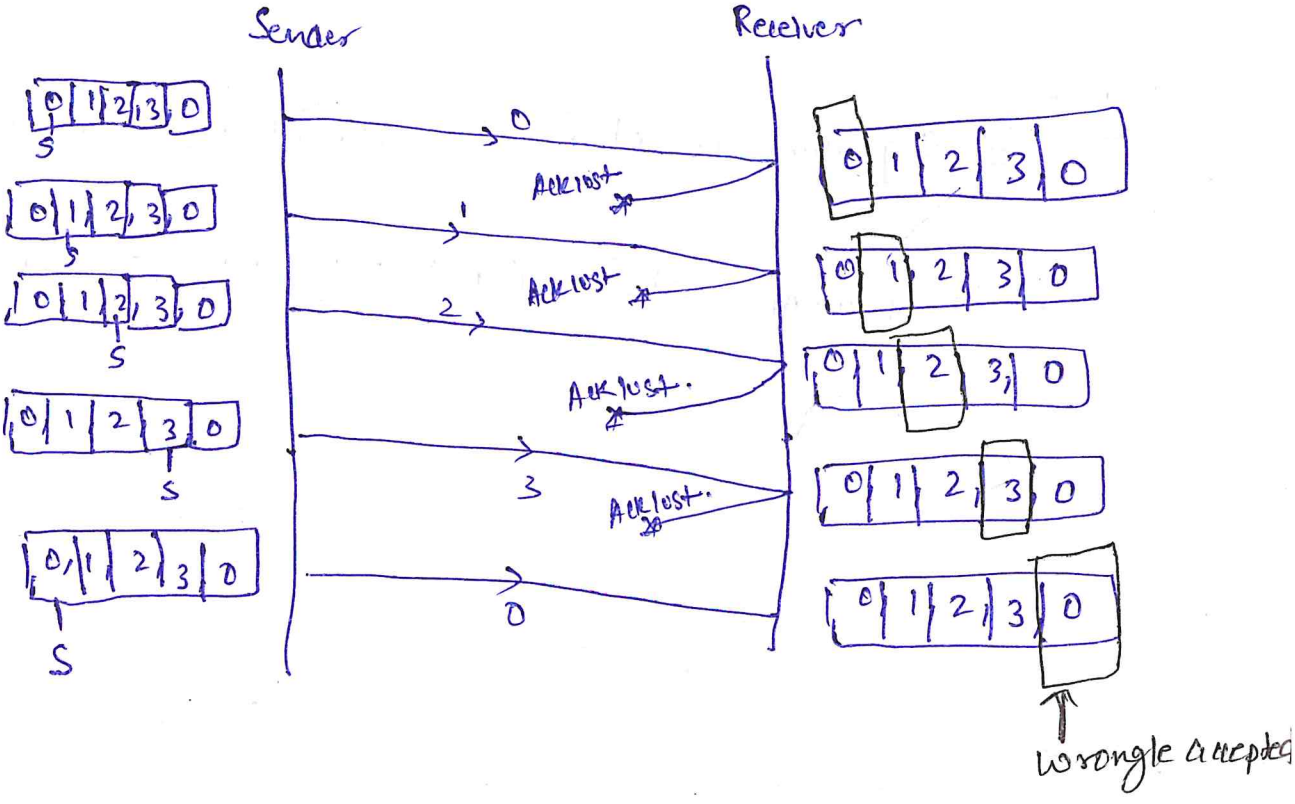
NOTE: [Receiver doesn't resend Ack in Go back N]

GBN : Sender-size window $= 2^m$ vs $2^m - 1$



Here receiver are waiting for 3 and sender is sending zero which is correctly discarded when we take $w.s \leq 2^m - 1$

Another case : $ws \leq 2^m$, Now check.



Here when we take $WS = 2^m$ then we are wrongly accepting 0, which should be discarded as sender is re-sending same window again.

Therefor | GBN $ws \leq 2^m - 1$ |

Que: GBN with WS = 3, Sender sends 10 frames, Every 5th frame is lost. max number of frames transmitted?

Soln→

1, 2, 3, 4, [5, 6, 7], 5, 6, 7, 8, 9, 7, 8, 9, 10, 9, 10

= 18 frames we send.

Concept →

Seq num needed in stop and wait = 2 = $w_s + w_R$ = 1 bit required

Seq num need in GBN = $w_s + 1$ = $w_s + w_R$

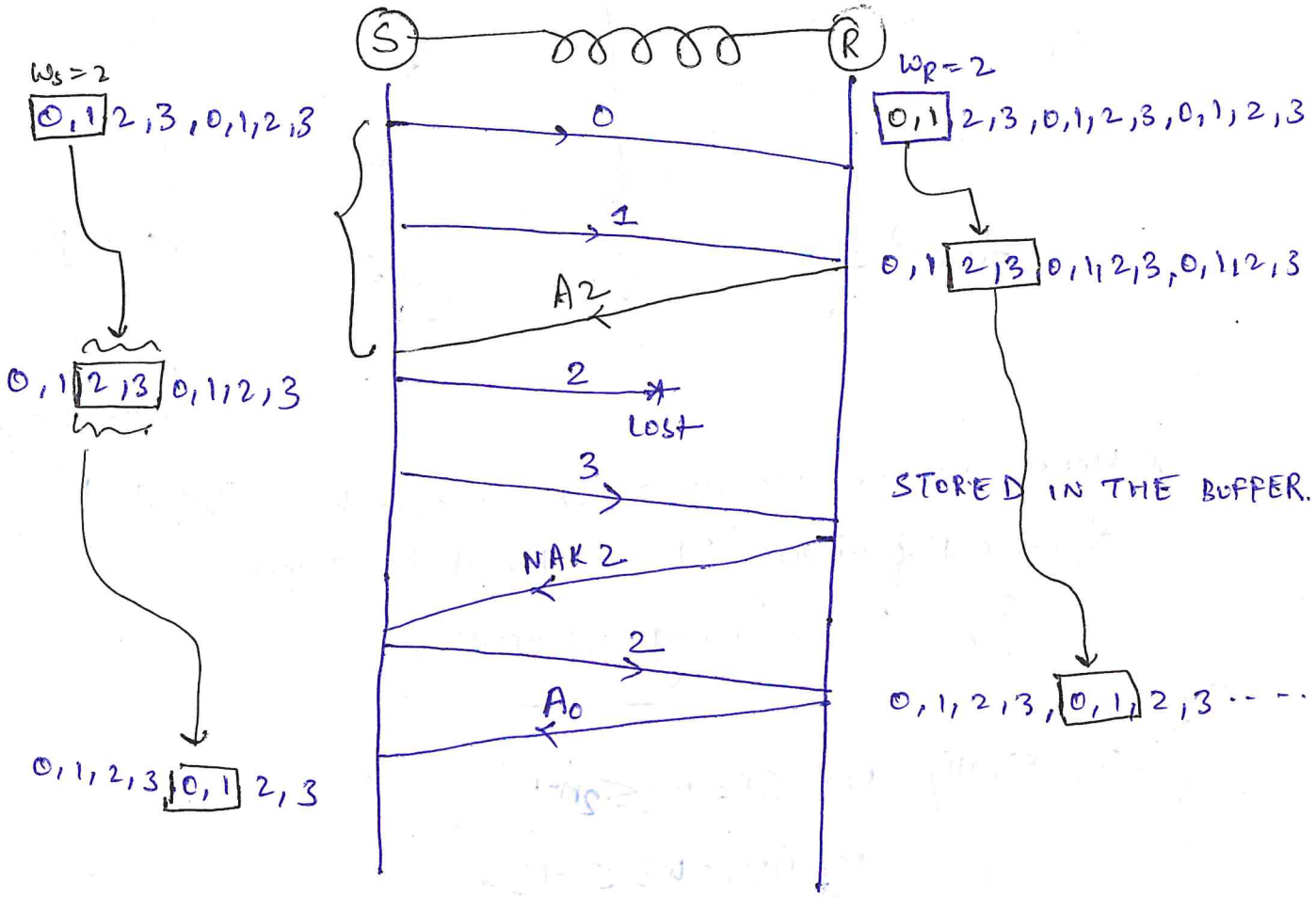
Selective Repeat SR (Sliding window)

→ Sender $w_S > 1$

→ Receiver $w_R = w_S > 1$

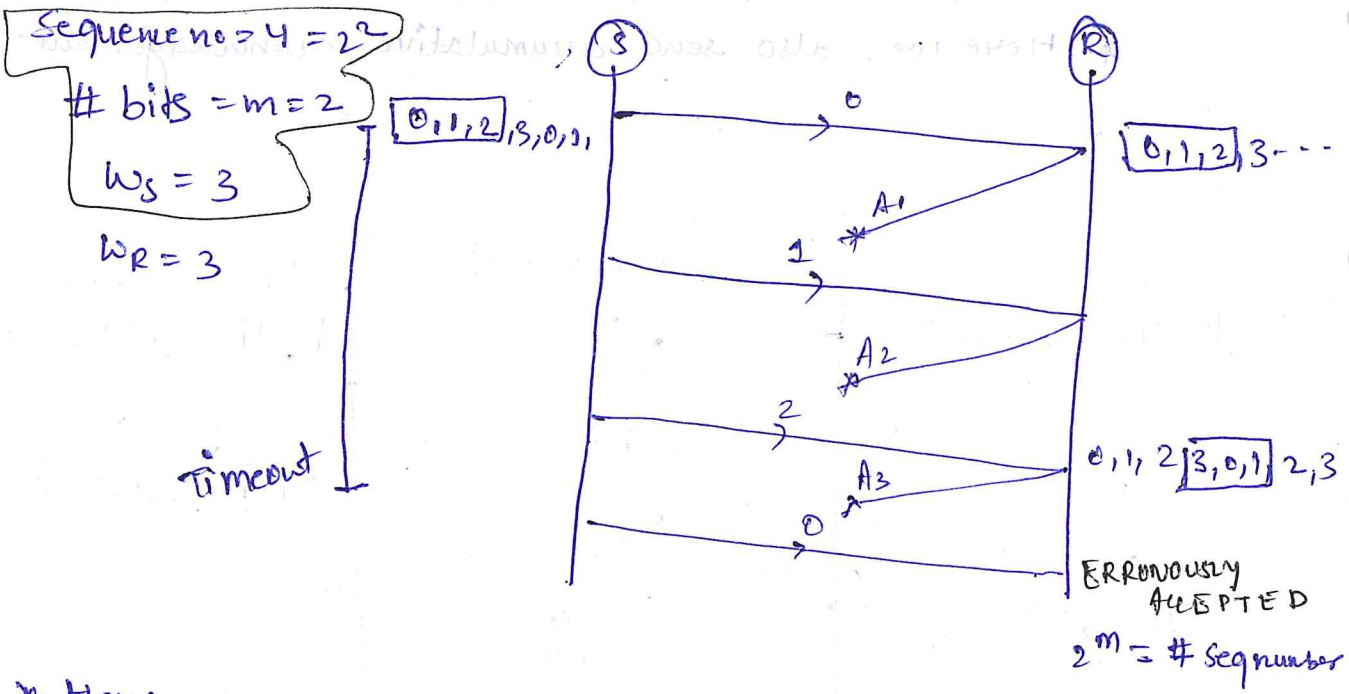
→ Re-send / Re transmission

- Implicit: R sends Ack for each good packet received. (→ Not accumulative)
- Explicit: R sends NAK for each corrupted / missing packet.
- * It supports both cumulative as well as individual Acks.



* Here we are sending a frame size of two. When any frame of the window is lost. We send the NAK frame regarding it. We need not to send the whole window again. We have to send only the frame which is lost.

Check for window size



2^m = # Seq number

* Here we are wrongly accept zero as part of second window. It should not be accepted

$w_S \le 2^{m-1} = \frac{2^m}{2}$

So, $w_S = 3$ [wrong assumption]

*

So, Finally for $SR = w_S \le 2^{m-1}$

For $GBN = w_S \le 2^m - 1$

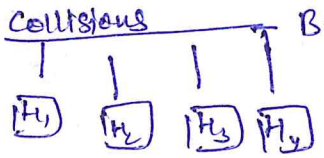
Q1. Concept →

	S&W	GBN	SR
η →	$1/(1+2a)$	$N/(1+2a)$	$N/(1+2a)$
No. of buffers →	2_S 2_R	$N+1$	$2N$
min no. of seq no. →	2 [0,1]	$N+1$	$2N$
Retransmissions →	1	N	(1) NAK
Implementation →	Simplest	Moderate	Complex.
BW →	low	high	Moderate.

* Selective repeat is used in wireless situations.
* GBN can be ideal you can use fibre optic.

Multiple / media - Access Protocols :-

- Channelization Protocols
  - FDMA
  - TDMA
  - CDMA
- Controlled-access protocols
  - Reservation
  - Polling
  - Token Passing
- Random access protocols
  - Aloha
  - SLOTted Aloha
  - CSMA/CD



→ Problem here is multiple host on the same channel, so there is chances of collision.

Channelization : Frequency division Multiple Access / Multiplexing

Freq: 100 MHz — 1000 MHz

100 - 199 MHz → H_1

200 - 299 MHz → H_2

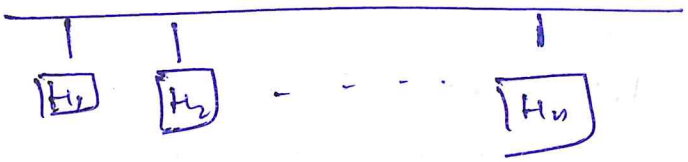
300 - 399 MHz → H_3

⋮

900 - 999 MHz → H_9

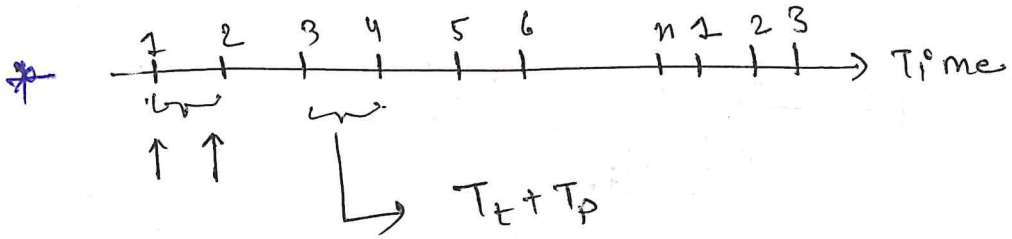
Here we are dividing the frequency such that no frequency will infere with each other

Time division multiple access:-



→ We are dividing the time into intervals.

→ In each interval we are giving turn to various hosts.



* assume all packets are of same size.

$$\eta = \frac{T_t}{T_t + T_p} = \frac{1}{1+a} \qquad \boxed{a = \frac{t_p}{t_t}}$$

$$\text{Throughput} = \eta \times B.$$

Drawback of TDM:-

* There are instances where some hosts or stations don't want to transmit data then that particular slot will be idle and no. transmission of data takes place

$$\eta = \frac{T_t \times 3}{4(T_t + T_p)}$$



↳ This is not sending

↳ Efficiency reduced

Que: $T_t = 1\,ms$

$T_p = 1\,ms$ ← TDMA

(a) $\eta = ?$ (b) $B = 4\,Mbps$

Throughput / available-BW =>

(c) If every station must needs 1Mbps bandwidth, No. of stations that can be connected.

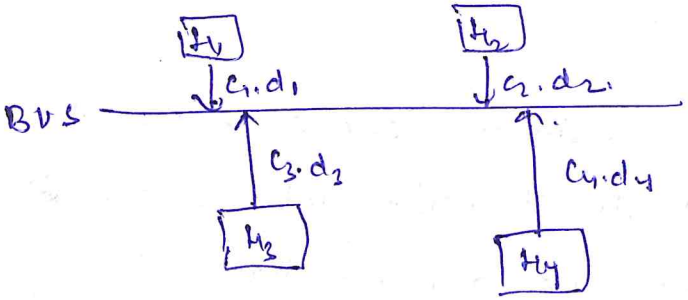
Soln:

(a) $$\eta = \frac{1}{1 + (T_p/t_t)} = \frac{1}{1+1} = 50\%$$

(b) $\eta \cdot B = 2\,Mbps$

(c) $$\frac{2 \times 10^6\ bps}{1 \times 10^3\ bps} = 2000$$

Channelization: Code division multiple access (CDMA)



$$c_1 \cdot d_1 + c_2 \cdot d_2 + c_3 d_3 + c_4 \cdot d_4 \cdot = \sum_{i=1}^{4} c_i d_i$$

→ Here we are putting data together in the medium.

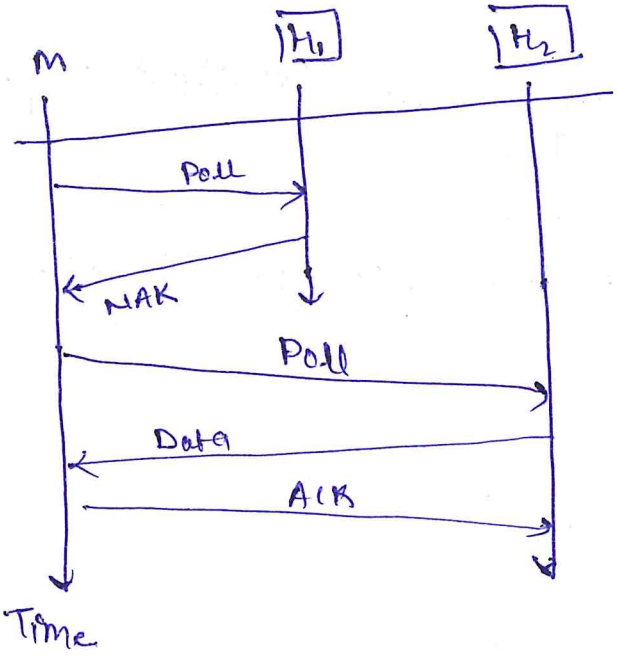
→ We can decouple data d_1, d_2, d_3, d_4 because we know what c_1, c_2, c_3, c_4 are.

CDMA → Used by Reliance Telecomm

Polling, Reservation, Token Passing

Polling: M: master / controller / primary

Common topology: STAR. → In case of polling



1, 2, 3 ... n ... 1, 2 ...
order

In a polling mechanism, master Poll the host 1. If host 1 sends the data it will send Data. Otherwise it send NAK frame and similarly master poll host 2 and so on.

→ Prioritizations

Let suppose we want to prioritize host 1

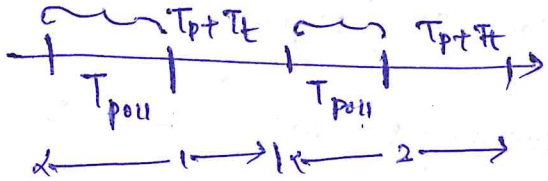
So, we will go like 1, 2, 1, 3, 1, 4 - - - -

After some time we are recalling one.

→ Band width utilisation is low in polling

→ Single point of failure means if master fails everything fails.

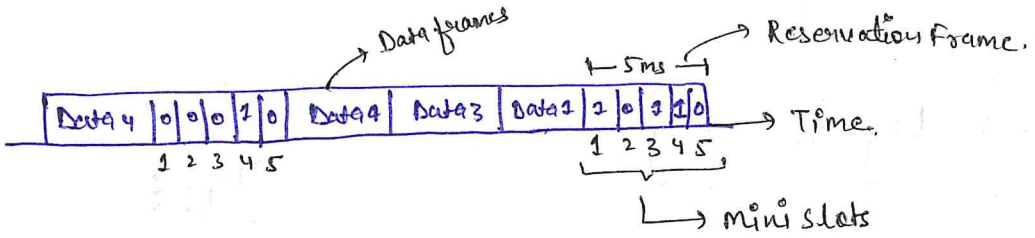
Timings →



$$\eta = \frac{T_t}{T_t + T_p + T_{poll}}$$, Throughput $= \eta \cdot B$

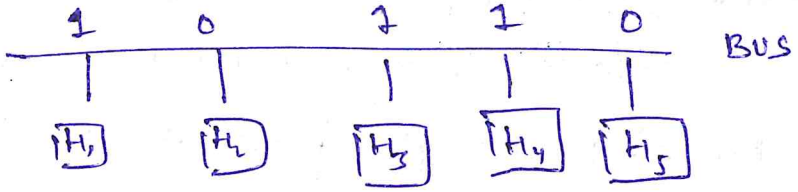
* Efficiency is lower, so BW is lower.

Reservation :-



In this approach we are putting the reservation frame on the channel.

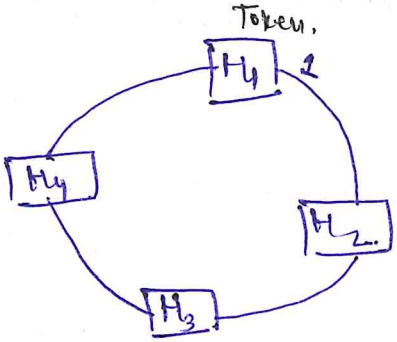
Each host we are providing 1ms to put their reservation bit



* Now we have default order H_1, H_2, H_3, H_4, H_5 and each host got to know who will send the data first.
* Accordingly H_1 place $Data_1$, Now H_3 place $Data_3$, and H_4 place $data_4$.
* When data-transfering is done we again put the reservation frame on the channel.

Token - Passing / Ring :-

→ Logical ring



→ Token : whoever have a token has a permission to transmit

→ No Ack is required because once the packet starts moving in the ring. The destination took that packet and again forwards in the ring. Once the packet reaches to ~~host~~ the source again. It got to know that transmission completed sucessfully.

→ Packet removal will be done by source itself.

→ When transmission is completed and source now can send the token on the ring.

→ Using ring topology broadcasting can be done very efficiently.

Concept → B-bit delay.

B : 1 sec → B bits
V : 1 sec → V mtrs.

Ex→ K-bit distance.?, Bandwidth = B bits/sec,

Soln→ 1 sec = B bits
1 bit = $1/B$ sec
K bits = K/B sec = $K/B \times V$ = KV/B metres.

Mathematics of token ring:

n - stations / hosts
d = distance (m)
v = velocity (m/s)

B : Bandwidth (bps)
K : delay @ each station in bits.

$$\text{ring Latency} = \left(\frac{d}{v} + \frac{n \cdot K}{B}\right) \text{sec}$$

$$= \left(\frac{dB}{v} + nK\right) \text{bits}$$

$$= T_p + \text{bit-delay}$$

↳ often time we take this as zero.

Token holding time :- (THT)

* The time each hosts hold the token.

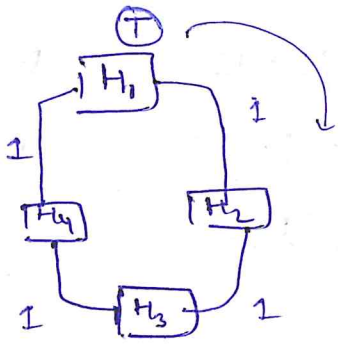
Cycle time: $\left(\frac{d}{v}\right)_{T_p} + (n * THT)$

$$\eta = \frac{\text{Useful time}}{\text{Cycle time}} = \frac{n \times T_t}{\left(\frac{d}{v} + n(THT)\right)}$$

assume: Each station sends exactly one frame.

THT (Delayed-token reinsertion)

→ Token held till it returns to the token holder



Only one packet at a time on the ring

$$THT_{DTR} = T_t + RL$$

$$\eta_{DTR} = \frac{\eta T_t}{\frac{d}{v} + \eta (T_t + RL)} = \frac{\eta T_t}{T_p + \eta \left(T_t + T_p + \frac{\eta K}{B}\right)}$$

Let K=0 bit delay

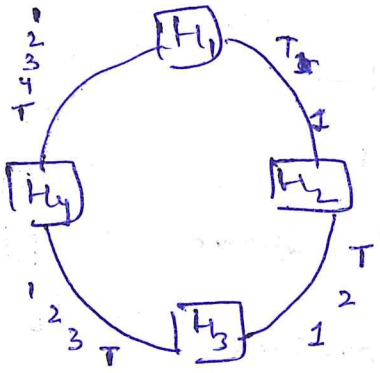
$$= a = \frac{T_p}{T_t}$$

$$\eta_{DTR} = \frac{1}{1 + \left(\frac{n+1}{n}\right) a}$$

=> Efficiency of Delayed token reinsertions

THT (Early-token-reinsertion)

$THT_{ETR} = T_t$ [Assuming bit-delay 0]



Multiple packets @ same time on the ring

$$\eta = \frac{n \cdot T_t}{T_p + \eta(T_t)}$$

$$\eta_{ETR} = \frac{1}{1 + a/n} > \eta_{DTR} =$$

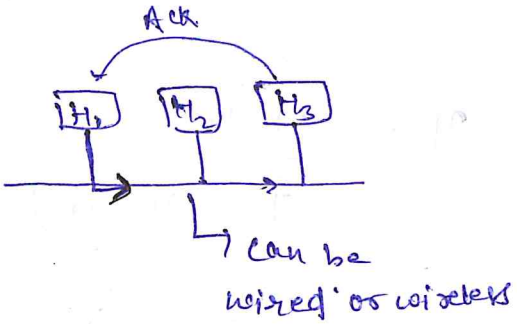
→ More efficiency and lower reliability.

* Here after sending a packet every host sending a token also onto the ring.

* In ETR corruption of data will be more as compared to DTR.

Random - Access - Protocol: Aloha. [Univ of Hawaii] → It is designed for wireless comm.

Pure Aloha :—



→ Just send the frame

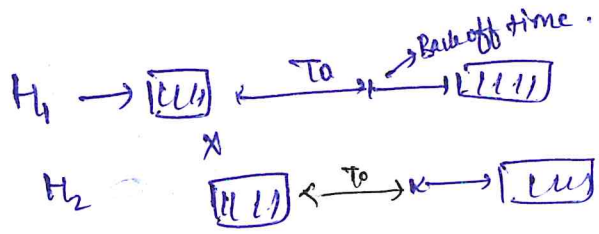
→ Ack from receiver

→ Collisions

→ Retransmit after random backoff time.

* Whenever you have a packet just send it without thinking of anything in pure aloha.

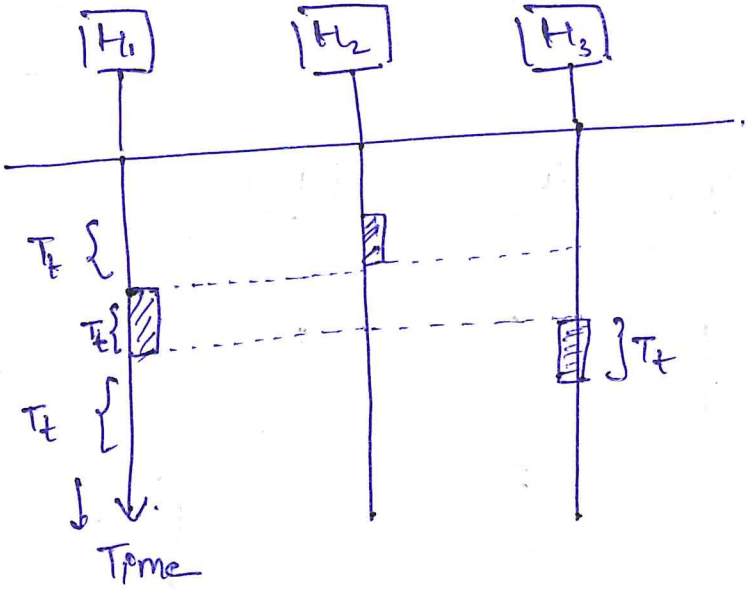
* chances of collisions more we are not waiting for anything.



* We are using back-off algorithm for the re-transmission.

Vulnerable time (T_v) = $2 \times T_t$

* The time where the possibility of collision.



* $\eta_{PA} = G_1 \cdot e^{-2G_1}$ [Poisson's distribution].

G_1 = No. of hosts that want to transmit

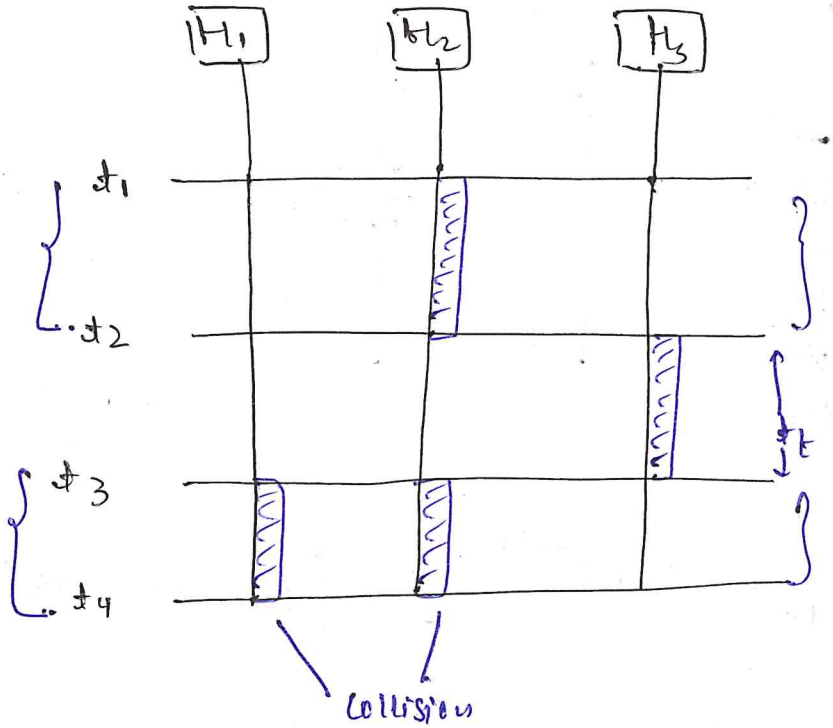
To maximize the efficiency we have to take derivative zero.

$$\frac{d\eta_{PA}}{dG_1} = 0$$

$G_1 = 1/2 \Rightarrow \eta_{PA}^{MAX} = 0.184 = 18.4\%$.

* Maxm efficiency of pure aloha is 18.4%.

Slotted Aloha:-



→ Vulnerable time = T_t

→ Here we are breaking the whole time axis every gap is equal to T_t.

→ Any station can transmit only start of the time slot.

*

$$\eta_{SA} = G e^{-G}$$

Max η_{SA} ⇒ $G = 1$

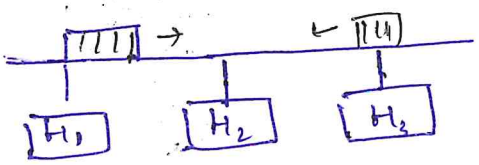
⇒ $\eta = 1/e = 0.368 = 36.8\%$.

CSMA : Carrier-sense multiple access :-

1 - persistent CSMA :

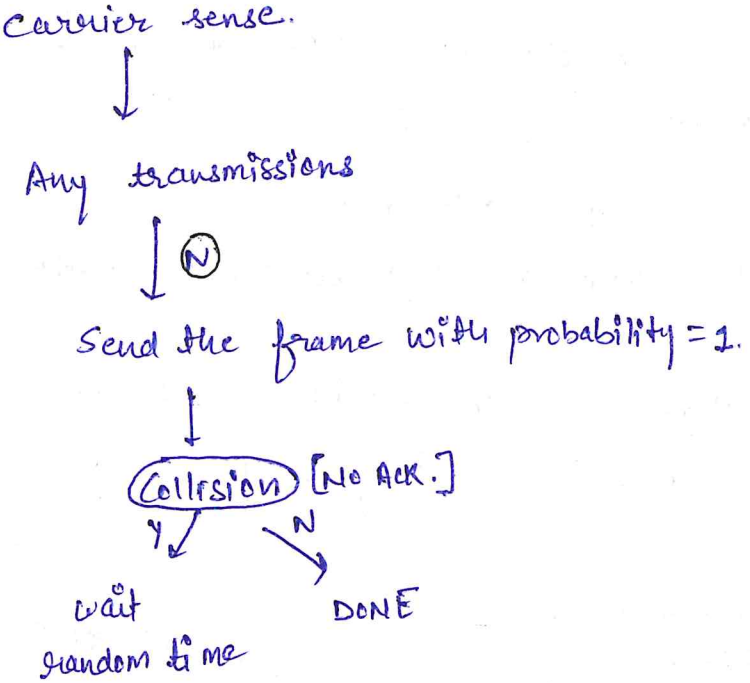
* collisions due to T_p
* Small T_p in real word (better than Aloha).

[Ethernet LAN]



→ It first senses the channel and whenever the channel is ideal. It will transmit with probability = 1.

→



Non-Persistent CSMA : —

→ Less greedy {whenever find any packet on the medium just wait for random amount of time}
→ longer delay than 1-persistent.

P-Persistent CSMA

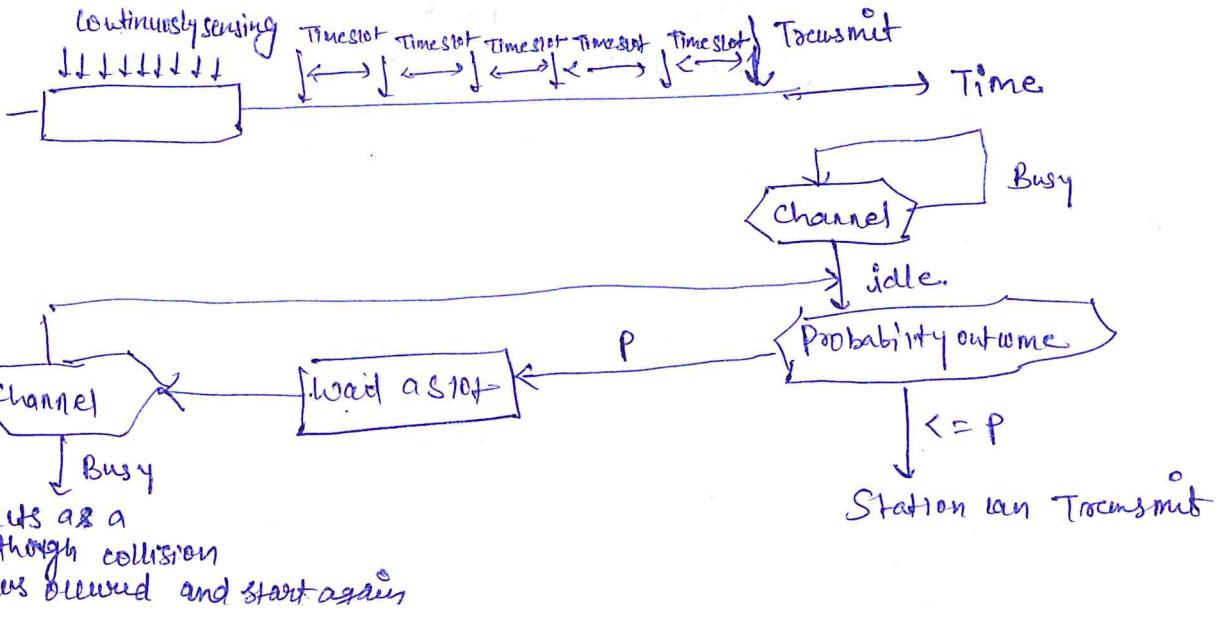
→ Used in wireless LANs
→ Slotting [concept of slotting]

Step-1 It keeps sensing the channel and busy.

Step-2 Once the channel is idle it generates the random variable $0 \leq r \leq 1$

if $r < p$ → Transmits
$r > p$ → wait for a slot → After sense the channel.

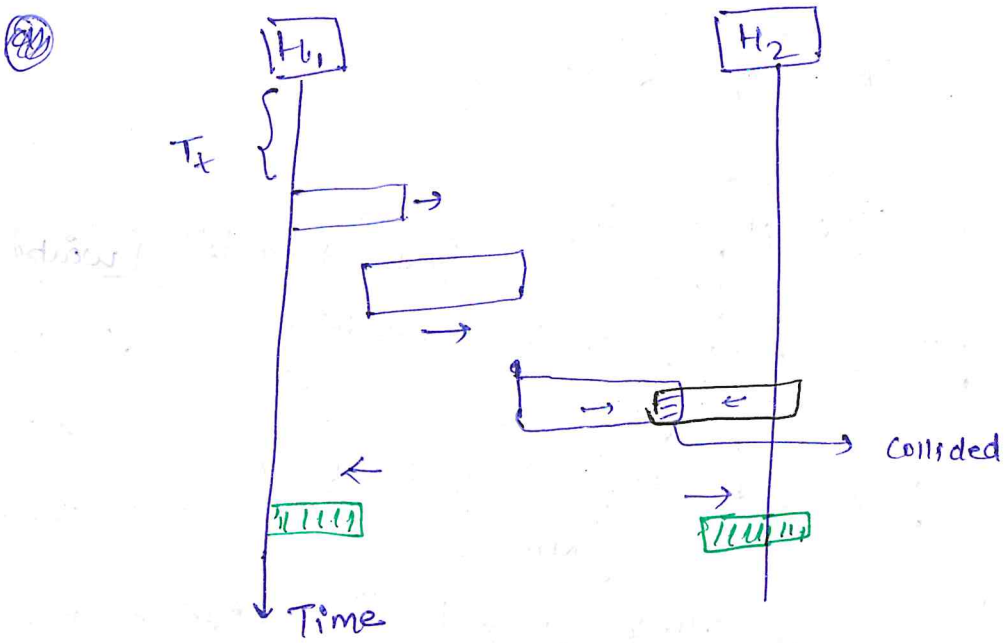
* When channel is idle it will transmit with probability p and not transmit with probability 1-p.



CSMA/CD :— (collision-detection)

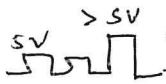
→ No-ACK-needed.

→ { Collision detection } media-dependent



* Here Packet of H_1 and H_2 get collided the H_1 and H_2 will not received the corrupted data.

(a) Compare sent and received state. } we use to find CD

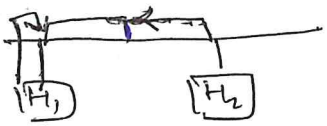
(b) high signal amplitude. ← dependent] → 

when any signal receive with more amplitude

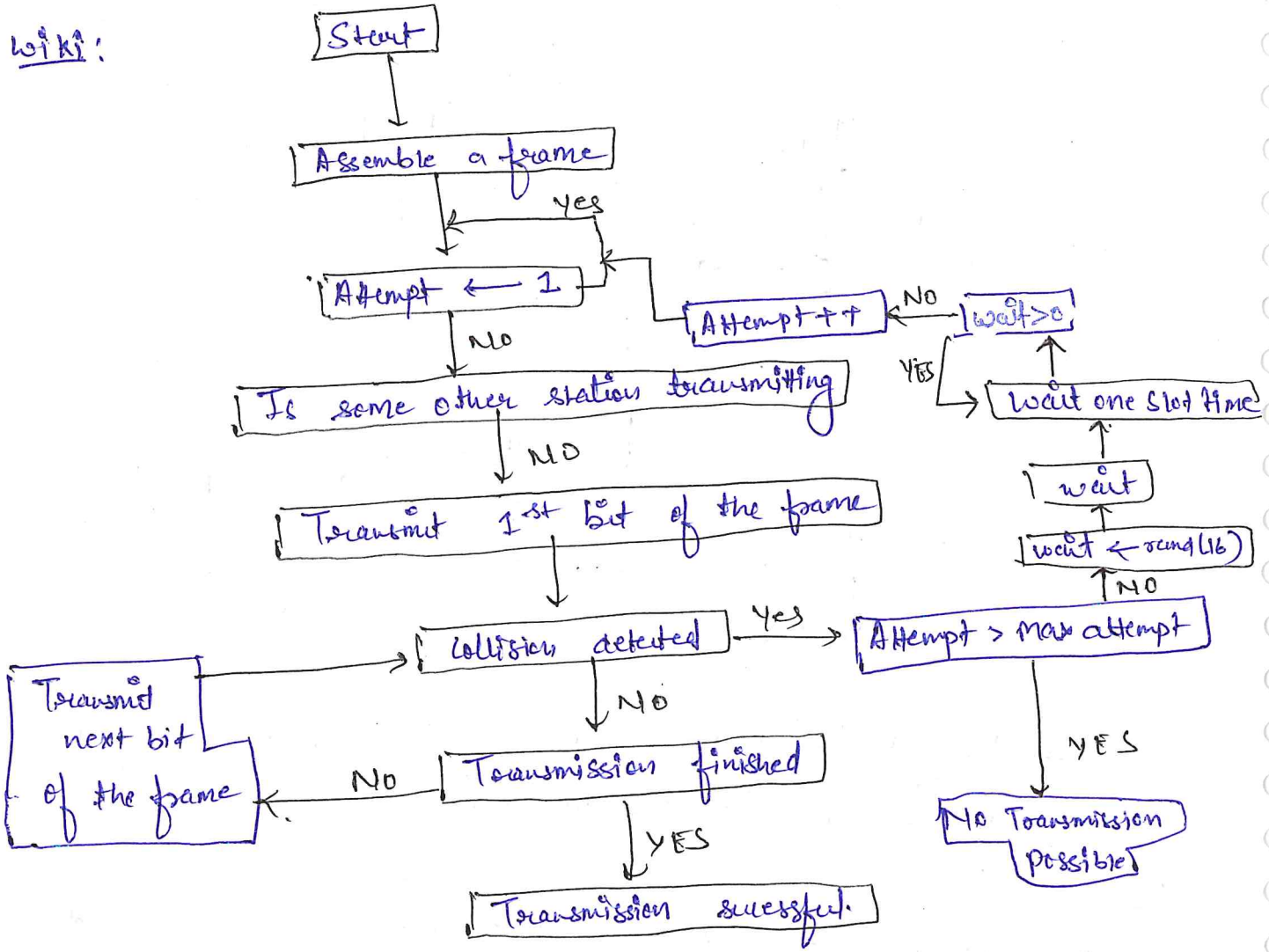
(c) receiving signal while still transmitting

↳ common approach.

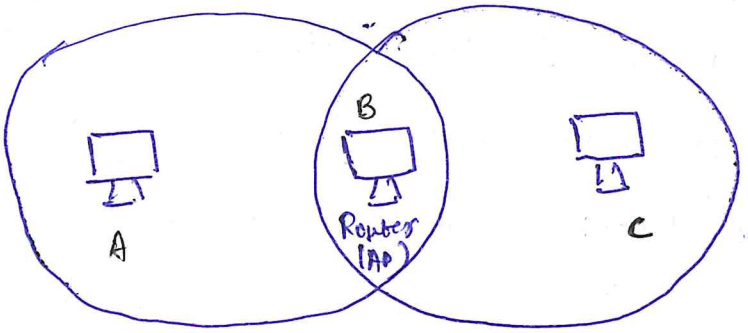
* We can analyze the signal when we are transmitting.



Wiki:



CSMA/CA : Collision avoidance [wireless LAN IEEE 802.11]

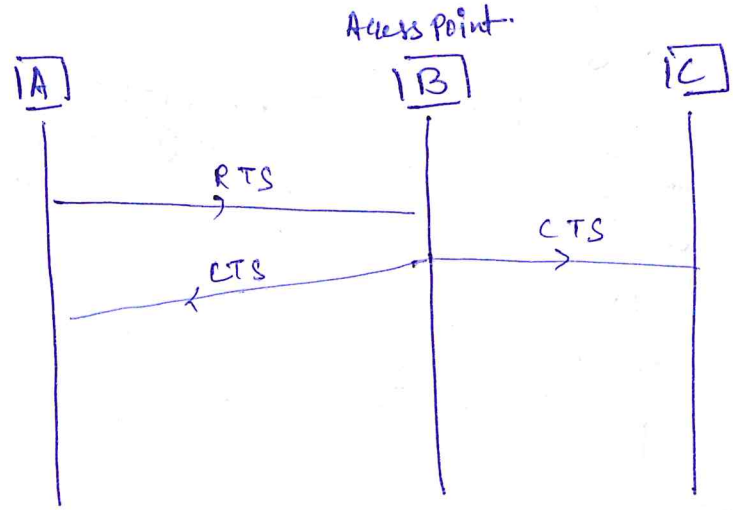


Hidden-node problem.

* A is not in range of C.
* When C is transmitting data, A doesn't know that C is transmitting.

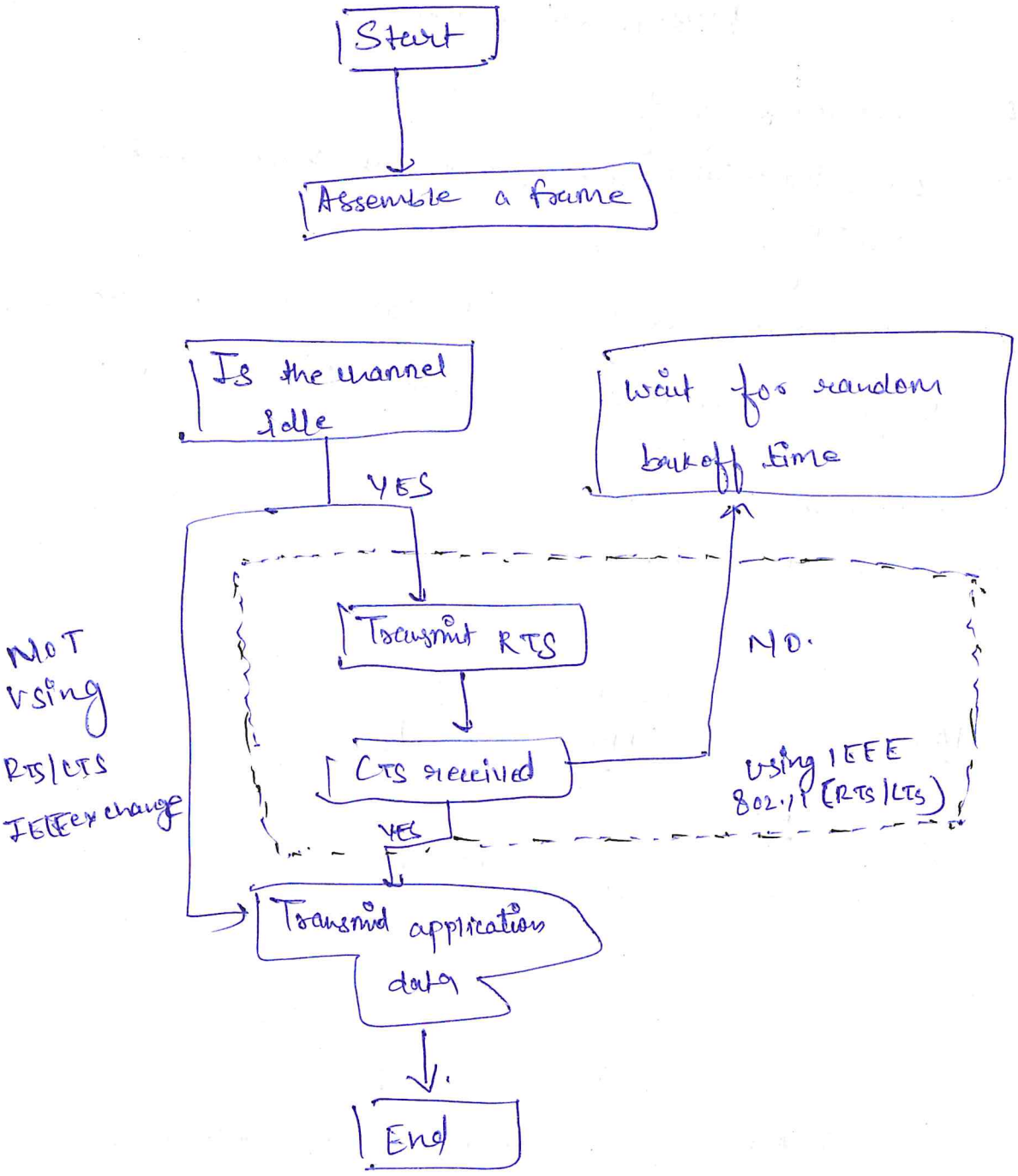
* Hidden node problem is very particular to csma' wireless.
* We use RTS/CTS exchange to solve the hidden node problem.

RTS → Request to send.
CTS → clear to send.



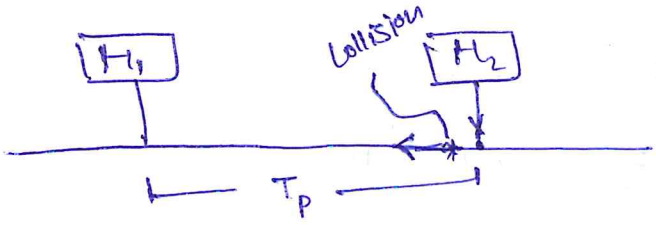
* Here node A send RTS request to node B. Node B send the CTS to A. So this A will

start transmitting the data to B. The CTS will also reach to node C. That is why C also knows that someone is transmitting to B and C will wait for some time before sending RTS to B. This solves the hidden node problem.



CSMA/CD analysis and efficiency:

CD: receving signal while transmitting



* Let suppose collision takes place just few dist near to host 2. The collision happens at propagation time.

* The collision signal reaches again (T_p time) takes to reach H_1.

To detect the collision.

$$T_t \geq 2T_p$$

$$\Rightarrow L/B \geq 2T_p$$

$$\Rightarrow L \geq 2T_p B$$

$$\Rightarrow \boxed{L \geq 2\frac{d}{V}B}$$

minimum size of frame to detect the collision.

Eg $T_p = 1ms$

$B_w = 1 Mbps$

min L for CD = ?

Solut $L \geq 2 T_p \cdot B$

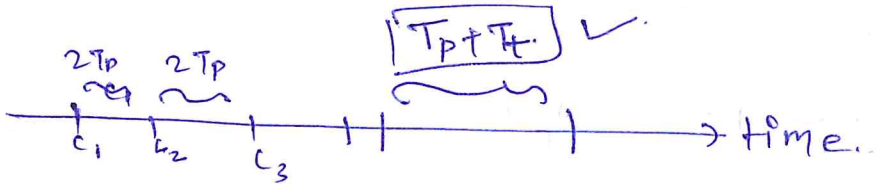
$\min L = 2 T_p \cdot B$

$\min L = 2 \times 10^6 \times 10^{-3}$

$= 2000 bits.$

$\eta_{CSMA/CD}$:- Time wasted due to one collision $= 2T_p$ (worst case)

$$\frac{UT}{TT} = \eta = \frac{T_t}{2T_p C + T_t + T_p}$$



* Here C_1, C_2, C_3 representing no. of collisions in the time axis.

η_t

Assume: only one station transmitting @ a time.

P = Probability that a stations transmits.

Probability of success $\leftarrow (P_S) = {}^{n}C_1 \cdot p \cdot (1-p)^{n-1} = n \cdot p \cdot (1-p)^{n-1} = f(p, n)$

Max P_S $\quad \frac{\partial P_S}{\partial p} = 0 \Rightarrow p = 1/n$

$\Rightarrow P_S^{max} = n \cdot 1/n \, (1-1/n)^{n-1}$

$\lim_{n \to \infty} P_S^{max} = 1/e.$

* If we have large number of stations the maximum probability of station $P_S^{max} = 1/e.$

Ex→ Number of times we have to try before successfully transmitting.

Soln→ $P_S^{max, n \to \infty} = 1/e.$

⇒ 1 success in e trails

Expected value of $e = e.$

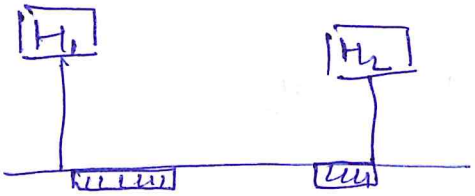
Now,

$$\eta_{(CSMA/CD)} = \frac{T_t}{2 \cdot e \cdot T_p + T_p + T_t} = \frac{1}{1 + 6.44a}$$

$$= \frac{1}{1 + 6.44\left(\frac{d}{v} \times \frac{B}{L}\right)}$$

$\uparrow \eta$ vs $L\uparrow$ or $\int n$ vs $d\uparrow$

Exponential back off Algorithm: →



One → How long to wait once we detect a collision?

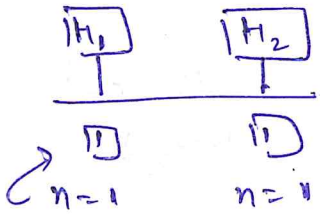
Waiting time = Back off time.

* Both the hosts should wait for random amount of time.

Waiting time = $W = K \times T_{slot}$

K: Choosen randomly.

Case - 1



$P(C_1) = 1$

* Both of the hosts collided for the first time

The random number H_1 will wait and H_2 will wait

$= [0 \text{ to } 2^n - 1] \Rightarrow$ For first time collision $n = 1$

H_1	H_2	
0	0	$\rightarrow C_2$ [second time collision]
0	1	$\rightarrow H_1$ success
1	0	$\rightarrow H_2$ success
1	1	$\rightarrow C_2$

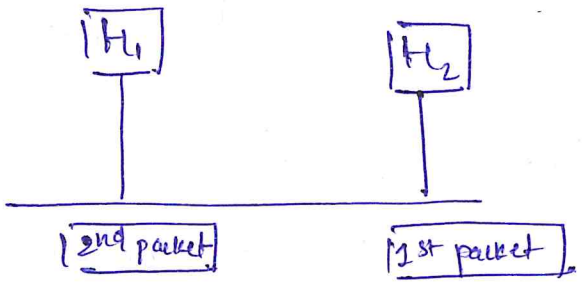
$P(C_2 | C_1)$ = probability of collision 2 after collision 1 $= \frac{2}{4} = \frac{1}{2} = 0.5$

$P(H_1 | C_1)$ = Host 1 successfully transmits after collision 1 $= \frac{1}{4} = 0.25$

$P(H_2 | C_2)$ = Host 1 successfully transmits after collision 1 $= \frac{1}{4} = 0.25$

Case-2 For the first time collision-1 takes place.
For the second time Host 1 successfully transmitted

Now, Host 1 is sending 2nd packet and Host 2 is sending 1st packet and collision 2 happens.



$n = 1$ $\quad$ $n = 2$ → because second time it get collided

$[0 \text{ to } 2^n - 1]$ $\quad$ $[0 \text{ to } 2^n - 1]$

$n = 1$ $\quad$ $n = 2$

$= (0, 1)$ $\quad$ $(0, 1, 2, 3)$

H_1	H_2
0	0 → C_3
0	1 → H_1 wins
0	2 → H_1
0	3 → H_1
1	0 → H_2
1	1 → C_3
1	2 → H_1
1	3 → H_1

$$P(H_1/C_2) = 5/8$$

$$P(H_2/C_2) = 1/8$$

$$P(C_3/C_2) = 2/8 = 1/4$$

* Using the Exponential back off the probability of 2nd collision after 1st collision will decrease.

Capture Effect:

* Winning hosts have the highest probability of winning again.

Solution: Use collision free algorithm : Token Passing

SOLVED PROBLEMS FROM GATE:

Que: The values of parameters for the stop and wait ARQ protocol are as given below:

Bit rate of the transmission channel = 1 Mbps

Propagation delay from sender to receiver = 0.75 ms.

Time to process a frame = 0.25 ms

Number of bytes in the information frame = 1980

Number of bytes in the acknowledge frame = 20

Number of overhead bytes in the information frame = 20

Assume that there are no transmission errors. Then the transmission efficiency (expressed in percentage) of the stop and wait ARQ protocol for the above parameters is ____ (correct to 2 decimal places).

Solution: $B_W = 1$ mbps $\quad T_P = 0.75$ ms

$T_{pr} = 0.25$ ms

$$\eta = \frac{UT}{\text{Total time}}$$

⇒ Number of bytes in information frame = 1980

⇒ Overhead bytes = 20

∴ actual data = $1980 - 20 = 1960$

$$TT = DS/B_W = \frac{1960 \times 8}{10^6} = \text{Useful time} = 15.68 \text{ ms}$$

Overhead bits $TT = 0.16$ ms.

$$\eta = \frac{UT}{TT} = \frac{15.68}{15.68 + 0.16 + 0.75 + 0.25 + 0.16 + 0.75 + 0.25}$$

- $15.68 + 0.16$ → TT including overhead
- $+0.75$ → PT
- $+0.25$ → Pr
- $+0.16$ → TT ACK
- $+0.75$ → PT ACK
- $+0.25$ → PT ACK

$$\eta = \frac{15.68}{\text{Total}} = 87\%$$

Que.2 A network has a data transmission bandwidth of 20×10^6 bits per second. It uses CSMA/CD in the MAC layer. The maximum signal propagation time from one node to another node is 40 microseconds. The minimum size of a frame in the network is ____ bytes.

Ans → $Bw = 20 \times 10^6$ b/s , $PT = 40 \times 10^{-6}$ s

$$T_t \geq 2T_p$$

$$\min T_t = 2T_p$$

$$L/B = 2 \times 40\ \mu s$$

$$\min L = 2 \times 40 \times 10^{-6} \times 20 \times 10^6$$

$$= 1600 \text{ bits} = 200 \text{ B}$$

Que Consider a LAN with four nodes S_1, S_2, S_3 and S_4. Time is divided into fixed size slots, and a node can begin its transmission only at the beginning of a slot. A collision is said to have occured if more than one node transmit in the same slot. The probabilities of generation of a frame in a time slot

by $S1, S_2, S_3$ and S_4 are 0.1, 0.2, 0.3 and 0.4, respectively. The probability of sending a frame in the first slot without any collision by any of these four station is ____.

(A) 0.462

(B) 0.711

(C) 0.5

(D) 0.652.

Ans: $\underbrace{0.1\,(0.8)\,(0.7)(0.6)}_{S_1} + \underbrace{0.9\,(0.2)\,(0.7)\,(0.6)}_{S_2}$

$\Rightarrow$

$+\,(0.9)(0.8)(0.3)(0.6) + \underbrace{(0.9)\,(0.8)\,(0.7)(0.4)}_{S_4}$

S_3

$\Rightarrow$ 0.462.

Que Consider a CSMA/CD network that transmits data at a rate of 100 Mbps (10^8 bits per second) over a 1 km (kilometre) cable with no repeaters. If the minimum frame size required for this network is 1250 bytes, what is the signal speed (km/sec) in the cable?

(A) 8000 (B) 10000 (C) 16000 (D) 20000

Sol. $d = 1\ km \quad B = 10^8\ bps \quad L = 1250 \times 8\ b$

$$T_t \geq 2 T_p$$

$$\Rightarrow \quad L/B \geq 2 \cdot d/v$$

$$\Rightarrow \quad \min L = 2\, d/v \times B$$

$$1250 \times 8 = 2 \times \frac{1}{v} \times 10^8\ bps$$

$$v = \frac{2 \times 10^8}{8 \times 1250} = \frac{10^8}{5 \times 10^3} = \frac{10^5}{5} = 2 \times 10^4 = 20,000$$

Que. A sender uses the stop and wait ARQ protocol for reliable transmission of frames. Frames are of size 1000 bytes and the transmission rate at the sender is 80 kbps (1 Kbps = 1000 bits/second). Size of an acknowledgement is 100 bytes and transmission rate at the receiver is 8 kbps. The one way propagation delay is 100 milli seconds. Assuming no frame is lost, the sender throughput is ____ bytes/second.

Sol:

$$L = 1000 \times 8\ b$$

$$B_S = 80 \times 10^3\ bps = 10\ KB$$

$$ACK \rightarrow 100 \times 8\ b = 100\ B$$

$$B_R = 8 \times 10^3\ bps = 10^3\ B/s$$

Throughput $= \eta \times B_s$

$$\eta = \frac{UT}{TT} = \frac{0.1}{0.1 + 0.1 + 0.1 + 0.1} = \frac{0.1}{0.4} = 1/4 =$$

$$= 1/4 \times 10{,}000 = 2500 \text{ B/s}$$

Que Consider a 128×10^3 b/sec satellite communication link with one-way propagation delay of 150 milli seconds. Selective retransmission (repeat) protocol is used on this link to send data with frame size of 1 Kilobyte. Neglect the transmission time of acknowledgement. The minimum number of bits required for the sequence number field to achieve 100% utilization is.

(A) 2 (B) 4 (C) 6 (D) 8

Ans :- $T_p = 150 \text{ ms}$ $l = 1 \text{ KB}$

$$\eta = \frac{W_s}{1 + 2a} = 100\%$$

$$T_t = \frac{1 \times 8 \times 10^3}{128 \times 10^3} \text{ sec}$$

$$W_s = 1 + 2a$$

$$\Rightarrow 1 + 2\frac{T_p}{T_t}$$

$$W_s = 5.687 \approx 6$$

$W_S = W_R = 6$

min sequence numbers $= 2N = 12$

For 12 sequence number = we need 4 bits.

Ans = 4.

Que Suppose that stop and wait protocol is used on a link with a bit rate of 64 Kilo bits per second and 20 miliseconds propagation delay. Assume that the transmission time for the acknowledgement and the processing time at nodes are negligible. Then the minimum frame size in bytes to achieve a link utilization of at least 50% —

(A) 160 (B) 320 (C) 640 (D) 220

Sol. $B = 64$ Kbps $T_p = 20$ ms

min $\eta = 50\% = 1/2 = \frac{UT}{TT}$

$= \frac{L/64\text{Kbps}}{L/64\text{Kbps} + 20\text{ms} + 20\text{ms}} = 1/2.$

$\Rightarrow 1/2\left(L/64 + 40\right) = L/64$

$L/128 + 20 = L/64 \Rightarrow L/128 = 20$

$\therefore L = \frac{128 \times 20}{8} = 320$ bytes.

Que: A Link has a transmission speed of 10^6 bits/sec. It uses data packets of size 1000 bytes each. Assume that the acknowledgement has negligible transmission delay, and that its propagation delay is same as the data propagation delay. Also assume that the processing delays at nodes are negligible. The efficiency of the stop and wait protocol in this setup is exactly 25%. The value of the one way propagation delay (in milliseconds) is

(A) 4 (B) 8 (C) 12 (D) 16

Sol: $B = 10^6$ bps $L = 1000 \times 8$ b $T_p = ?$

$$T_t = \frac{1000 \times 8}{10^6} = 8 \text{ msec}$$

$$\eta = \frac{1}{4} = \frac{UT}{TT} = \frac{8ms}{8 + T_p + T_p}$$

$$\frac{1}{4} = \frac{8}{8 + 2T_p} \Rightarrow 2T_p = 24$$

$$T_p = 12$$

Que. Consider a network connecting two systems located 8000 Kilometers apart. The bandwidth of the network is 500×10^6 b/s. The propagation speed of the media is 4×10^6 meters per second. It is needed to design a Go-back N sliding window protocol for this network The average packet size is 10^7 bits. The network to be used to its full capacity. Assume that that the processing delays at nodes are negligible. Then, the minimum size in bits of the sequence number field has to be.

(A) 2 (B) 4 (C) 8 (D) 16

Ans: $d = 8000$ km, $B = 500 \times 10^6$ bps $V = 4 \times 10^6$ mps

$l = 10^7$ bits

$$\eta = 1 = \frac{W_s}{1+2a} \Rightarrow 1 + \frac{2T_p}{T_t} = W_s$$

$$T_p = \frac{d}{v} = 2, \quad T_t = l/B = \frac{10^7}{50 \times 10^7} =$$

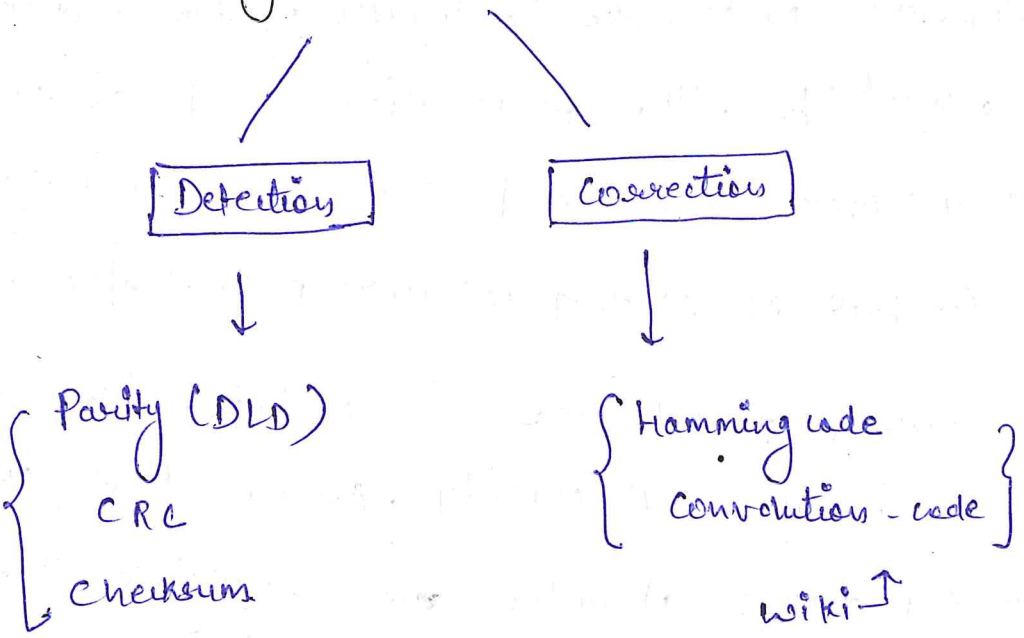
$$W_s = 1 + (2 \times 2 \times 50) = 201$$

No. of sequence no $= W_s + 1$

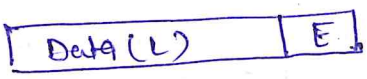
$= 202$

No. of bits $= \log_2 202 = 8$ bits.

Error-handling in MAC Layer.

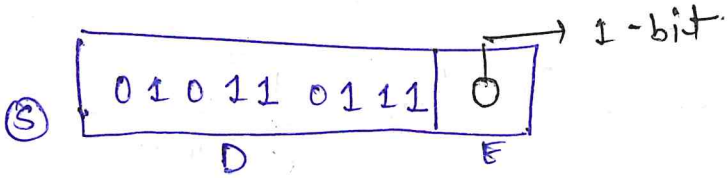


* We are going to focus on error detection much.



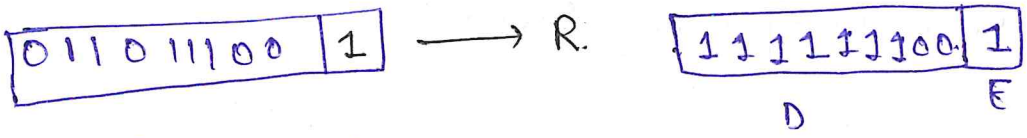
E → Error detection bits E << L

PARITY :— Even zero parity



Here we want to make the number of zeroes to be even.

* If Receiver find odd zeros than there is error.
  If receiver find even zeros than not sure there is error.

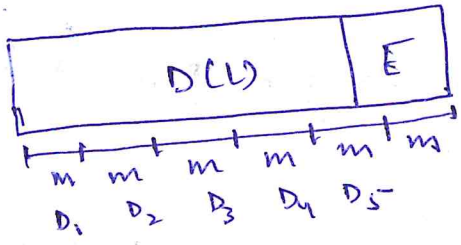


even zero's

→ Here at the receiver end if two bit change happen. The receiver think it is of even parity and accepted the corrupted data.

* cannot handle / detect errors in 2-bits or more.

Checksum :-



$$E = f(D_1, D_2, \cdots D_5)$$

Ist strategy : function is addition

$$E = D_1 + D_2 + D_3 + \cdots D_5$$

IInd strategy :

$$E = D_1 + D_2 + \cdots D_5 + c(D_1 + D_2 + \cdots D_5)$$

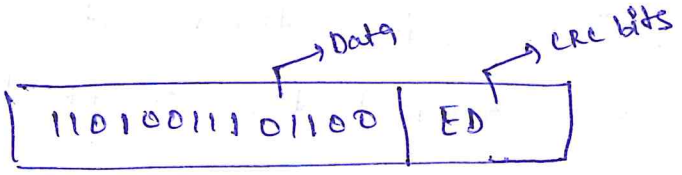
* Here in this method we are adding error detection bits in the last of data. We are performing some operation on data to get the error detection bits. The same thing is repeated by the receiver when it gets the data.

Cylic Redundancy check (CRC)

→ Uses division

→ based on Galois fields (GF)

Ex →



$x^3 + x + 1$ → Galois field polynomial / CRC polynomial.

* Sender and receiver both agree upon the CRC bits.

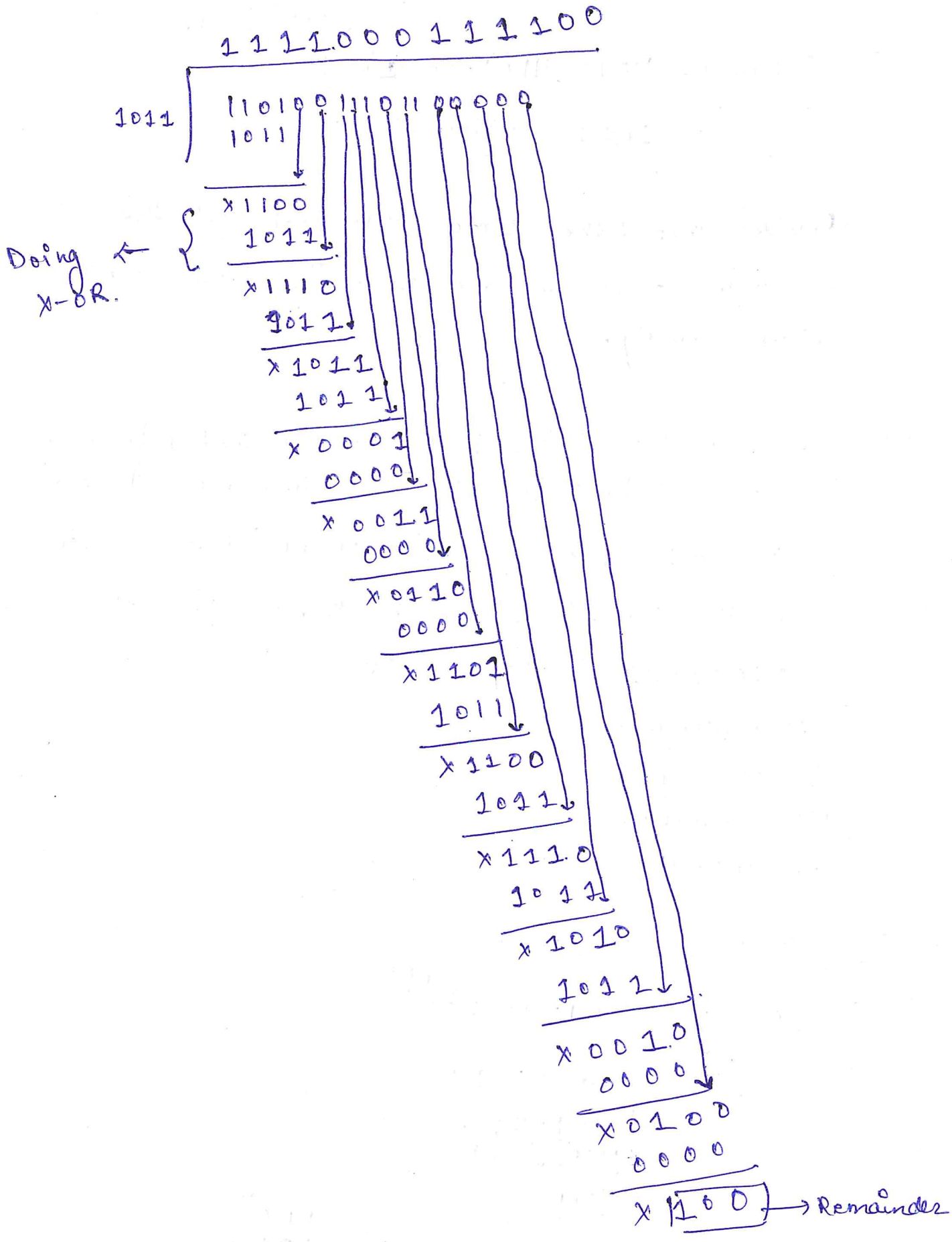
* If CRC bit = 3 than largest exponent is also three.

So, $x^3 \times 1 + x^2 \cdot 0 + x \cdot 1 + 1 \cdot 1 = x^3 + x + 1$

Divisor = 1011.

* Since we are using three bits CRC we have to add three zero's in data.

1101001110 1100 → Three additional bits



* Now we replace the remainder with three zero's and send to the sender.

So, Sent data = 1101001110111 00 → CRC bits calculated.

1011 → divisor

Receiver repeat the same step and get the remainder zeros in the last. This means data sent correctly.

Que: A computer network uses polynomial over GF(2) for error checking with 8 bits as information bits and uses x^3+x+1 as the generator polynomial to generate the check bits. In this network, the message 01011011 is transmitted as

(A) 010110110

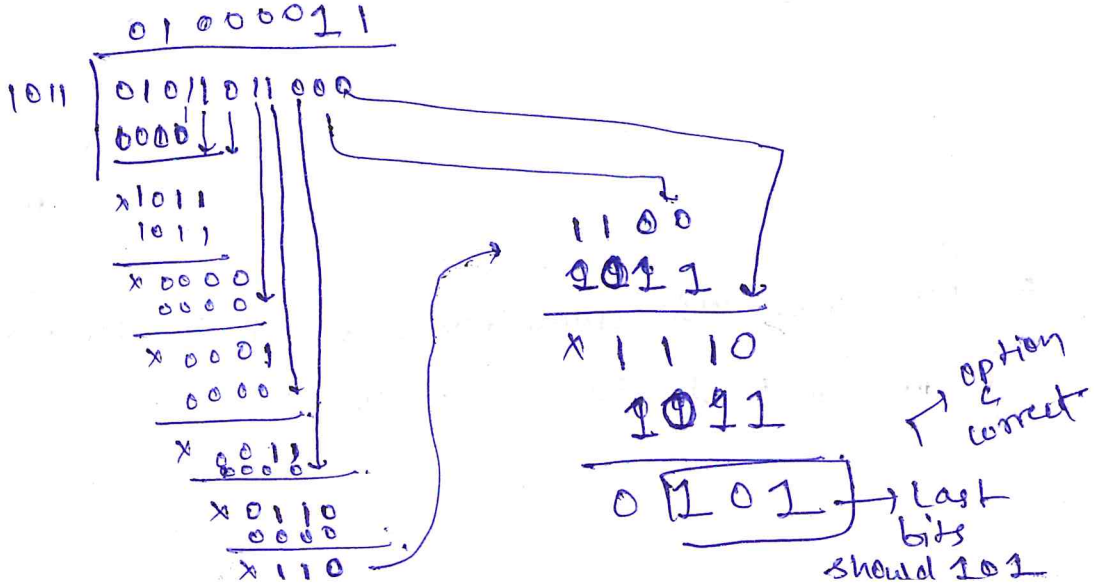
(B) 010110111

(C) 010110111

(D) 010110111 00

Sol: x^3+x+1 = 1011

message = 01011011 000 ← CRC



Framing

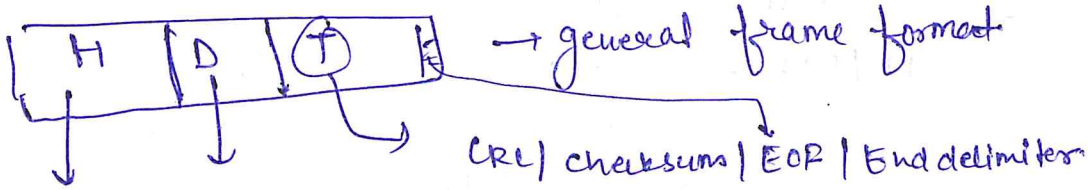
Network - layer (Packet)

↓

DLL (FRAME)

↓

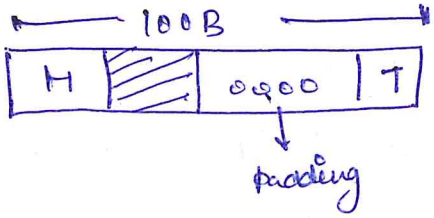
Physical layer



Frame no.
Size
protocol
SOF
Sender mac address
Receiver mac address

Packet from network layer.

Fixed size framing



* It wastes the space some time because we always need to follow some size.

Variable-sized framing

→ length (Ethernet)

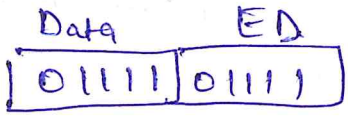
→ EOF (Token-ring) | end-delimiter

Problem → What if end-delimiter occurs in the data part?

Data = 01111

ED / EOF = 01111

Sender →



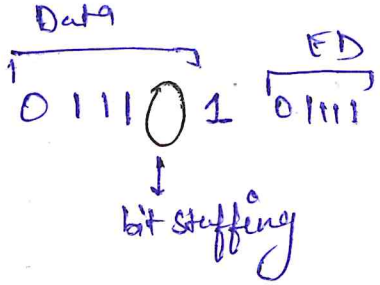
So, receiver can think 'Data as ED because both ED and data are same.

To handle this problem we are using bit stuffing.

* In bit stuffing we are stuffing 0/1 based on the ED

ED = 01111 ⇒ we make a rule that after every three one's we will stuff '0' in data part.

So, Now data =



Ex→ A-bit-stuffing based framing protocol uses 8-bit delimiter pattern of 01111110. If the output string after stuffing is 011111100101, then the input string is?

Ans:

ED → 0 111 111 0] → We will stuff zero after 5 one.

Remove ↓

O/p after stuffing : 0 1111 1 00 1 0 1

↳ Remove zero after 5 one.

Actual data = 0 11111 0 1 0 1

LAN- Protocols : Ethernet, Token ring, WLAN.

(I) Ethernet : IEEE 802.3

→ Constantly evolving (wiki): 2.94 Mbps, 10Mbps, 100 Mbps, 1Gbps, 25Gbps, ...

→ Bus Topology

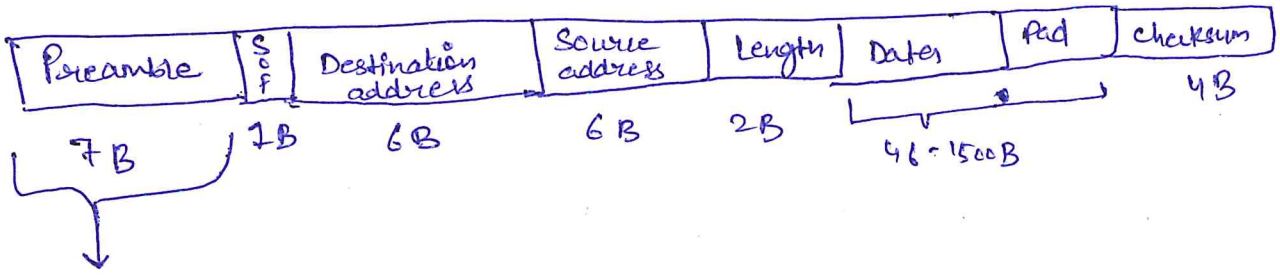
→ CSMA/CD ⇒ No ACK. [MAI]

→ Manchester encoding [Physical-Layer]

→ Expo. backoff algorithm.

* Twisted cable / Co-axial cable / fibre optical

FRAME - format (802.3).



10 1010 1011

* Preamble and sof are generally not considered as part of frame.
* Preamble and sof are added by the physical layer.
* Preamble is sending a clock signal to synchronize the receiver.
* Sof is the starting of frame.

* Destination and source address are the MAC address which are of 6B = 48 bit

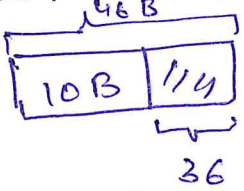
6B = 48 bits = 12 Nibbles

1 Nibble = 4bits → 0 .... 9 . A ... F

* MAC address are represent using a hexadecimal symbol.
* Length field tells us about the length of frame excluding preamble and SOF.

* Max size of frame = 1518 B
  Min Size of frame = 64 B

Concept → If your minimum data size is 10B in the frame. Then you have to add 36B padding to make it to 46 B.



* We are limiting the maximum size of frame is 1518 because. if we didn't set the upper limit then all other users will get blocked.

→ Protocol efficiency = $\frac{\text{Pay-load size}}{\text{Packet size}}$ = $\frac{1500}{1518}$ = 98.81 %.

→ condition for CD : $L \geq 2 \cdot \frac{d}{v} \cdot B \Leftarrow TT \geq 2T_p$.

→ collision vs real time applications

↳ max time to transmit successfully

→ real time applications [video calls, Youtube live]

In real time application we expect less errors.

Token - Ring (802.5)

→ Encoding : Diff manchester.

→ Token-passing ⇒ No collisions, No ACK.

→ 4mbps, 16mbps, 1 Gbps

→ Prioritization and server - client architectures.

* In token ring we need a logical ring topology.

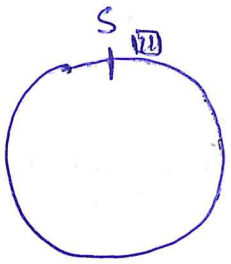
* If any host has the higher priority for holding the token then the token holding time for that host is more as compared to other hosts.

* In token ring there is no collisions and no Ack so we know exact time for the frame transmission. It is feasible to design real-time application using token ring.

Dis-advantages:-

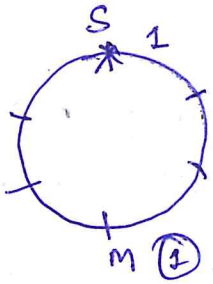
→ Source-down → orphan packet.
→ corrupted packet
→ Monitor bit

* If our source is down then who will remove that packet from the ring.?



These packet are called the orphan packet.

Solution:



→ After sending a packet, the sender went down.
→ The monitor node first time marked that packet as one.
→ If that packet come again to the monitor then monitor will removed that packet from the ring considered as orphan.

→ Token problems → captured, Lost or corrupted

* Monitor keeps track of maximum token return time
If token doesn't return in that time then monitor created the new token.

→ Destination → down or busy

↳ Available and copy bits used by the destination if destination is down and busy.

* The destination re-set the monitor bit for negating the monitor effect.

→ Link → down/cut.

→ Monitor → down.

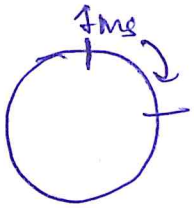
↳ monitor keep sending Active monitor presence at regular Interval to all host [Heart beat msg].

Que. Token-holding time = 1ms

B = 16 Mbps

Max frame size = ?

Ans.

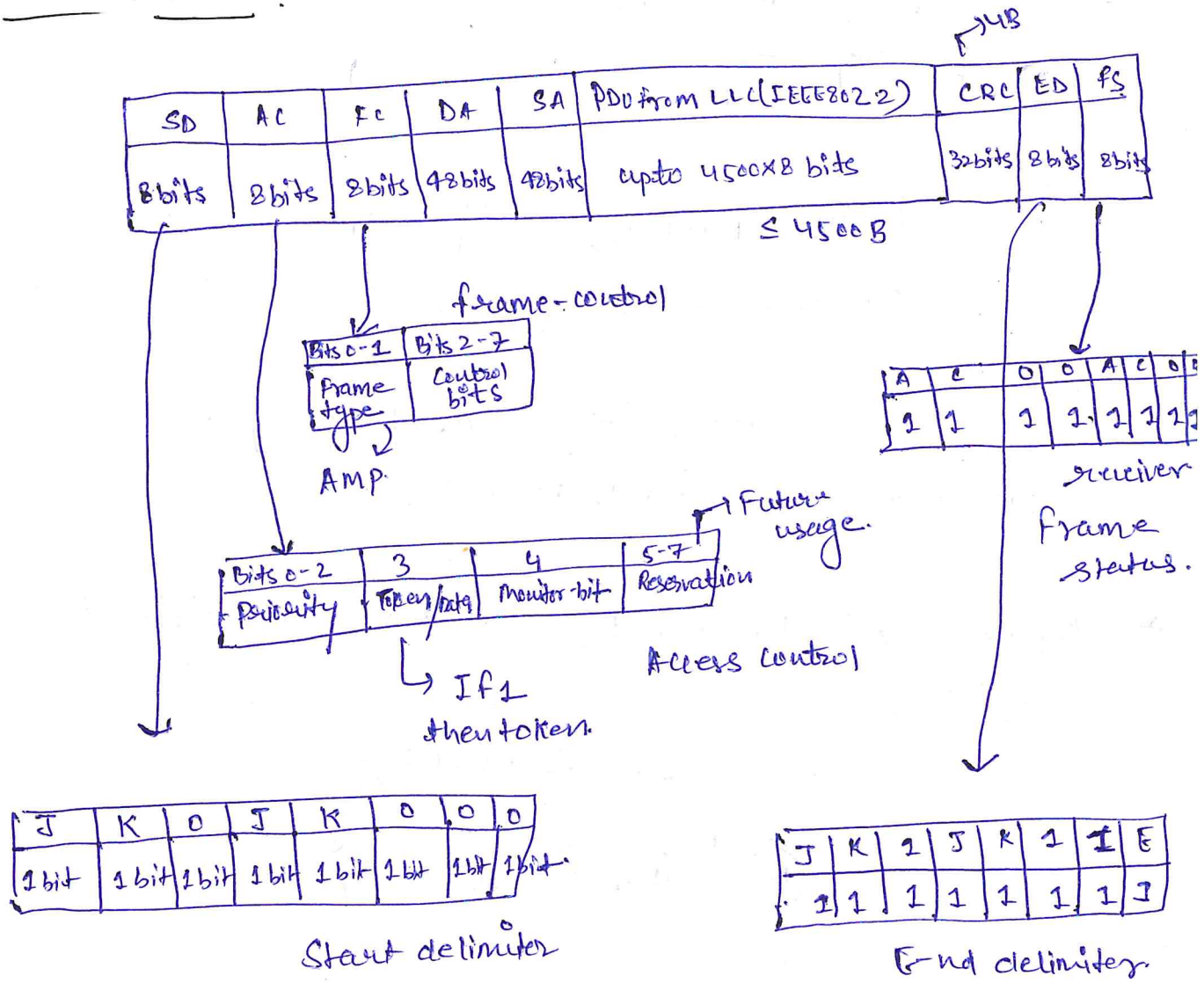


while holding a token how many bit you can transfer.

$1 s = 16 \times 10^6$ bits

$1 ms = 16 \times 10^6 \times 10^{-3} = 16 \times 10^3$ bits.

Frame Format:



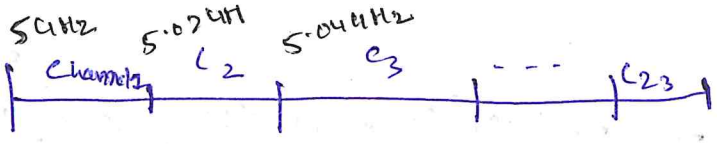
802.11 [WLAN] Wireless LAN.

→ CSMA/CA (ACK, RTS/CTS)

→ Wireless : 2.4GHz, 5GHz.

→ Medium is air [Radio] [EM waves]

5GHz means → 23 non-overlapping half-duplex channels.



1GHz = 1000MHz.

20MHz = 0.02GHz.

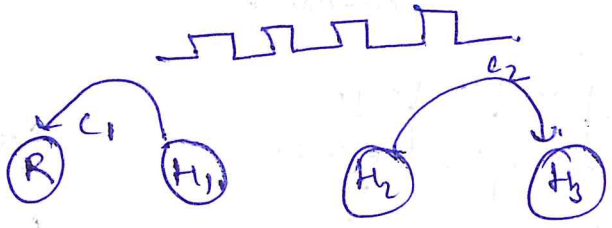
C1 : 5.502 GHz

C2 : 5.02 - 5.04GHz

⋮

* Here we are dividing the 5GHz frequency in 0.02 GHz. Like that 23 channels possible.

* By using this H1 and H2 can transfer on different channels

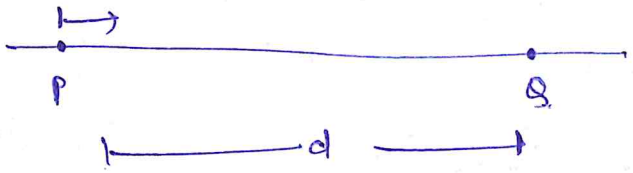


Que Consider a simple communication system where multiple nodes are connected by a shared broadcast medium (like ethernet or wireless). The nodes in the system use the following carrier-sense the mediums access protocol. A node that receives a packet to transmit will carrier sense the medium for 5 units of time. If the node does not detect any other transmission in this duration, its start Transmitting the packet in the next time unit. If the node detects another transmission, it waits until this other transmission finishes, and then begins to carrier sense for 5 time units again. Once they start to transmit, nodes do not perform any collision detection and continue transmission even if collission occurs. All transmissions last for 20 units of time. Assume that the transmission signal travels at the speed of 10 meters per unit time in the medium.

Assume that the system has two nodes P and Q, located at a distance d meters from each other. P starts transmitting a packet a time $t=0$ after sucessfully Completing its carrier-sense phase. Node Q has a packet to transmit at time $t=0$ and begins to carrier sense the medium.

The maximum distance d (in meters, rounded to the closest integer) that allows Q to sucessfully avoid a Collision between its proposed transmission and P's on going transmission is ?

Sol.



Carrier sense = 5 units

T_f = 20 sec

V = 10 m/s

The time taken a frame to reach Q is $= \frac{d}{10}$

⇒ To avoid collision our frame from P should reach to Q.

⇒ $\frac{d}{10} \leq 5$

$d \leq 50$ ⇒ max d = 50.

Que. In an Ethernet local area network which one of the following statements are true ?

(A) A station stops to sense the channel once it starts transmitting a frame.

(B) The purpose for the jamming signal is to pad the frames that are smaller than the minimum frame size.

(C) A station continues to transmit the packet even after the collision is detected.

(D) The exponential backoff mechanism reduces the probability of collision on retransmissions.

Sol: I. for 2.4GHz we have = 3 channels
for 5GHz we have = 5 channels
→ True.

II. There is no collision detection we have collision avoidance.
→ false.

III. True, Unicast H_i —— H_j
One node to one node = Unicast.

Ans → I and III only.